

Turán densities of complete 4-graphs via flag algebras on seven vertices

Bruno Mérola Corrêa
brunomerola@gmail.com

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Abstract

For $r \geq 3$ the Turán density $\pi(K_t^{(r)})$ of the complete r -graph is not known for any $t > r$. Among 4-graphs the only case with a plausible conjecture is $K_5^{(4)}$: Giraud’s construction gives $\pi(K_5^{(4)}) \geq 11/16$, and the best published upper bound is 0.73628. We prove $\pi(K_5^{(4)}) < 0.69160$, which removes 91.6% of the gap. We also determine the codegree-squared density $\sigma(K_5^{(4)}) = 31/64$, as conjectured by Balogh, Clemen and Lidický, with a stability theorem: near-extremal $K_5^{(4)}$ -free graphs are close to complements of balanced Giraud systems. We prove $\pi(K_6^{(4)}) < 0.86018$ and $\pi(K_7^{(4)}) < 0.93524$ (previously about 0.8946 and 0.9444), $\pi(K_5^{4-}) < 0.44701$ and $\pi(K_6^{4-}) < 0.78055$ for $K_5^{(4)}$ and $K_6^{(4)}$ minus an edge, and bounds for three 5-graphs and nine threshold families of 4-graphs. We improve the codegree-squared bounds for six 3-graphs, among them J_4 and $K_5^<$, give such bounds for four further 4-graphs, and show that the value conjectured for $\sigma(J_4)$ rests on an arithmetic slip. Each bound is a plain flag-algebra certificate on seven vertices with exact rational entries; its value is the exact minimum of an explicit function over all admissible 7-vertex graphs, computed by exhaustive search in two independent programs. On six vertices the same method gives exactly de Caen’s bounds for $K_5^{(4)}$ and $K_6^{(4)}$ and nothing for $K_7^{(4)}$, which may explain why we found no earlier flag-algebra bound on the Turán density of a complete 4-graph. An exact dual point shows that no plain seven-vertex certificate can improve our bound for $K_5^{(4)}$ by more than $2.2 \cdot 10^{-7}$. Consequences are asymptotic lower bounds for lottery numbers and covering insertion codes; the quadruples of every Giraud-type construction cost at least $7/16$ to cover fractionally by k -sets, sharp for $k = 6$.

1 Introduction

For an r -graph F , the Turán number $\text{ex}(n, F)$ is the largest number of edges of an n -vertex r -graph with no copy of F , and the Turán density is $\pi(F) = \lim_{n \rightarrow \infty} \text{ex}(n, F) / \binom{n}{r}$. Turán [45] determined $\pi(K_t^{(2)}) = 1 - 1/(t-1)$. For $r \geq 3$ no value $\pi(K_t^{(r)})$ with $t > r$ is known, and Erdős offered \$500 for any one of them and \$1000 for all [13]; see the surveys [41, 24]. The best-studied case is the tetrahedron $K_4^{(3)}$, whose density Turán conjectured to be $5/9$. Razborov’s flag algebras [38] gave $\pi(K_4^{(3)}) \leq 0.561666$ [39], Baber [2] gave 0.5615, and Jeong, Park, Im, Lee and Yang [21] recently reached 0.557789 (arXiv v2) with a seven-vertex certificate, found by cutting planes and column generation and checked in Lean.

The 4-uniform case. For 4-graphs the analogous problem is $K_5^{(4)}$. Giraud [18] constructed 4-graphs of density $5/16 + o(1)$ in which every five vertices span an edge (Definition 2); their complements are $K_5^{(4)}$ -free of density $11/16 + o(1)$. Sidorenko conjectured that $11/16$ is the answer [41, Conjecture 7] and wrote: “The only case with $r > 3$, when there exists a plausible

conjecture on the asymptotic behavior of Turán numbers, is $k = 5$, $r = 4$ ” [41, Section 8]. The upper bounds have moved little. The general bound $\pi(K_{r+1}^{(r)}) \leq 1 - 1/r$ of Sidorenko and de Caen [40, 10] gives $3/4$. Sidorenko proved 0.74912 [40] (as reported in [26]), and Lu and Zhao derived 0.74439 in the concluding remarks of [26], without giving the proof of one ingredient (their value $R = 13$). Markström [27] determined $\text{ex}(n, K_5^{(4)})$ for $n \leq 16$ by computer search, found that Giraud’s construction attains it for every $n \leq 16$ (uniquely for $13 \leq n \leq 16$), and proved $\text{ex}(17, K_5^{(4)}) \leq 1753$, that is $T(17, 5, 4) \geq 627$, which gives $\pi(K_5^{(4)}) \leq 1753/2380 \approx 0.73655$. Sidorenko [42, Section 2] observed that integer rounding at $n = 18$ gives $751/1020 \approx 0.736275$. That bound, or Markström’s, is still quoted as the best one in recent work [3, 34, 17]. The open interval for $\pi(K_5^{(4)})$ was $[0.6875, 0.73628]$, of width 0.0488 .

For $K_6^{(4)}$ and $K_7^{(4)}$ we know of no conjectured value. De Caen’s bound $\pi(K_t^{(r)}) \leq 1 - 1/\binom{t-1}{r-1}$ [10] gives $9/10$ and $19/20$. The small Turán numbers in Sidorenko’s tables [42], together with repeated averaging $T(n, p, r) \geq \lceil nT(n-1, p, r)/(n-r) \rceil$, give about 0.8946 and 0.9444 ; here this is slightly stronger than de Caen’s averaging bound $t(p, r) \geq T(s, p, r)/(\binom{s}{r} - 1)$ [11, Theorem 3]. Sidorenko’s constructions [42] show $\pi(K_6^{(4)}) > 0.8458$ and $\pi(K_7^{(4)}) \geq 4677/5120 \approx 0.9135$.

Why flag algebras had not been used. Flag algebras have been applied to 4-graphs before. Balogh, Clemen and Lidický [3, Section 8.2] used them for the codegree-squared density of $K_5^{(4)}$, obtaining $\sigma(K_5^{(4)}) \leq 0.55241250$ against the lower bound $31/64$ from Giraud’s construction, and they quote Markström’s bound for $\pi(K_5^{(4)})$. Chen, Hou, Liu, Zhang and Zhao [7] ran 4-graph flag algebras on six vertices for tight cycles. We found no flag-algebra bound on the Turán density of any complete 4-graph. Theorem 9 below suggests why: on six vertices, plain flag algebras (sums of squares of flag densities only, without stationarity or other additional constraints) return exactly de Caen’s bounds $3/4$ and $9/10$ for $K_5^{(4)}$ and $K_6^{(4)}$, and nothing for $K_7^{(4)}$. The plain method needs seven vertices. A 4-graph on seven vertices is the same object as a 3-graph on seven vertices (replace every edge by its complement), and there are 7013320 of these up to isomorphism. The semidefinite program has Gram blocks of order up to 1024 and several million linear constraints. We solve it with the cutting-plane and column-generation scheme of Jeong et al. [21], in an independent implementation, and we evaluate the rounded certificate exactly on every admissible seven-vertex graph.

Results. Theorem 4 gives the bounds for $K_5^{(4)}$, $K_6^{(4)}$, $K_7^{(4)}$, K_5^{4-} and K_6^{4-} , and Theorem 5 the three 5-graph bounds; Table 1 compares them with the literature. Theorem 8 gives nine threshold families (Table 2). Theorem 6 proves the conjecture $\sigma(K_5^{(4)}) = 31/64$ of Balogh, Clemen and Lidický [3, Conjecture 8.2 of arXiv v3]. Its certificate is tight on the seven-vertex subgraphs of Giraud’s construction. Theorem 7 gives stability with linear dependence and no removal lemma, using a characterisation of Giraud systems by their seven-vertex subgraphs (Theorem 16). Section 5 improves the upper bounds for ten further codegree-squared densities (Table 5; for the four 4-graphs among them we found no published bound) and shows that the value 0.28 conjectured for J_4 in [3] rests on an arithmetic slip: the construction behind the conjecture gives $16/57$, and the plain seven-vertex method, whose optimum we locate in an interval of width $4.2 \cdot 10^{-7}$, stops at least $1.5 \cdot 10^{-6}$ above it. Theorem 9 is the six-vertex barrier, and Theorem 31 shows that our seven-vertex bound for $K_5^{(4)}$ is within $2.2 \cdot 10^{-7}$ of the best that the plain seven-vertex method can give. Section 8 derives the consequences: lower bounds for lottery numbers (Corollary 34), for covering insertion codes (Corollary 35), and a theorem with a short human proof, Theorem 36, on how expensive Giraud’s systems are to cover. It also gives three upper bounds for asymptotic lottery numbers, by constructions and

Kahn's theorem [22] (Proposition 45); the one for $(6, 4, 5)$ shows that Theorem 36 is sharp for $k = 6$.

How the proofs are checked. The bounds are computer-assisted. Each certificate is a list of exact rational positive semidefinite matrices (for the 4-graph bounds of Theorem 4, integer matrices A with $Q = A^\top A/M^2$); Proposition 11 turns it into a lower bound b , and b is the exact minimum of an explicit rational function over all admissible 7-vertex 4-graphs. Two programs computed that minimum independently, over all labelled one-vertex extensions of the 6-vertex classes (up to $1.64 \cdot 10^8$ graphs), without canonical forms (Section 3.6). The certificates are not formally verified: no proof assistant has checked them.

Organisation. Section 2 states the results. Section 3 describes the method and proves Theorem 6 (Section 3.8), Section 4 proves the stability theorems, Section 5 treats further codegree-squared densities, Section 6 proves the six-vertex barrier and the seven-vertex limits, and Section 7 collects numerical observations on the seven-vertex optimum. Section 8 gives the consequences and Section 9 the data. Appendix A lists the parameters of the certificates, and Appendix B collects the notation.

2 Definitions and results

An r -graph G is a set $V(G)$ of vertices with a set $E(G)$ of r -subsets, the edges; $e(G) = |E(G)|$ and $d(G) = e(G)/\binom{|V(G)|}{r}$. For $S \subseteq V(G)$, $G[S]$ is the induced subgraph.

Definition 1 (Turán systems). Let $p > r$ and $\lambda \geq 1$. An r -graph G is (p, λ) -admissible if every p -subset of $V(G)$ spans at least λ edges; for $\lambda = 1$ we say p -admissible, and G is then a *Turán* (n, p, r) -system. Let $T_\lambda(n, p, r)$ be the least number of edges of a (p, λ) -admissible r -graph on n vertices, and $T(n, p, r) = T_1(n, p, r)$.

Admissibility is hereditary: every induced subgraph of an admissible graph is admissible. By averaging over the $(n-1)$ -subsets, $T_\lambda(n, p, r)/\binom{n}{r}$ is non-decreasing in n ([23, Theorem 1 and Corollary 1] for $\lambda = 1$), so the limit $t_\lambda(p, r) = \lim_n T_\lambda(n, p, r)/\binom{n}{r}$ exists; $t(p, r) = t_1(p, r)$ is the *Turán density* in Sidorenko's normalisation [41, 42]. An r -graph is p -admissible exactly when its complement in $K_n^{(r)}$ is $K_p^{(r)}$ -free, so

$$\pi(K_p^{(r)}) = 1 - t(p, r).$$

More generally, let H_k^r be the r -graph on $r+1$ vertices with k edges (it is unique up to isomorphism, since an r -subset of an $(r+1)$ -set is determined by the missing vertex). An r -graph is H_k^r -free iff every $(r+1)$ -set spans at most $k-1$ edges, iff its complement spans at least $r+2-k$ edges in every $(r+1)$ -set. Hence $\pi(H_k^r) = 1 - t_{r+2-k}(r+1, r)$. We write $K_5^{4-} = H_4^4$, the 4-graph on five vertices with four edges; then $\pi(K_5^{4-}) = 1 - t_2(5, 4)$. Also $K_6^{(5)} = H_6^5$. Finally K_6^{4-} is $K_6^{(4)}$ minus one edge: a 4-graph contains K_6^{4-} iff some six vertices span at least 14 edges, so $\pi(K_6^{4-}) = 1 - t_2(6, 4)$.

Giraud systems.

Definition 2 (Giraud systems). Let $V = A \cup B$ be a partition, one part possibly empty, and M a 0/1 matrix indexed by $A \times B$. The *Giraud system* $\Gamma = \Gamma(A, B, M)$ is the 4-graph on V whose edges are all 4-subsets of A , all 4-subsets of B , and the *cross* 4-sets $\{a, a', b, b'\}$ ($a, a' \in A, b, b' \in B$) whose minor $M_{ab} + M_{ab'} + M_{a'b} + M_{a'b'}$ is even.

Giraud's construction [18] is a Giraud system with a specific matrix and equal parts; it is a Turán $(n, 5, 4)$ -system whose complement is $K_5^{(4)}$ -free. With equal parts the internal 4-sets have density $1/8 + o(1)$ and the cross ones $3/8 + o(1)$, so every Giraud system with equal parts in which a fraction $1/2 + o(1)$ of the minors are even, for instance one with a uniformly random matrix, has density $5/16 + o(1)$; we use the name Giraud's construction for these balanced systems too. Definition 2 is the form described by Sidorenko [42, Section 2], with arbitrary M and arbitrary part sizes. At uniformity 4 it is also exactly Construction 2 of [41], the generalisation by de Caen, Kreher and Wiseman [12].

Lemma 3. *Every Giraud system is odd: every five vertices span 1, 3 or 5 edges. In particular it is a Turán $(n, 5, 4)$ -system. If $|A| = \alpha n$, $|B| = \beta n$, $s = \alpha\beta$ and f is the fraction of even minors among the pairs of pairs, then $d(\Gamma) = 1 - 4s + 2s^2 + 6s^2f + O(1/n)$, and $f \geq \frac{1}{2} - \frac{1}{2(\max(|A|, |B|) - 1)}$ when both parts have at least two points.*

Proof. A five-set inside one part spans five edges. A five-set with four points in one part spans exactly one, the internal 4-set. A five-set $\{a_1, a_2, a_3, b, b'\}$ spans only the three cross sets $\{a_i, a_j, b, b'\}$; with $\rho_i = M_{a_i b} + M_{a_i b'}$ the three minors are $\rho_i + \rho_j$, whose sum is even, so one or three of them are even. The case of three points in B is symmetric.

For the density, $\alpha^4 + \beta^4 = 1 - 4s + 2s^2$ and $\binom{\alpha n}{2} \binom{\beta n}{2} = 6s^2 \binom{n}{4} + O(n^3)$. For f , assume $|B| \geq |A|$ (otherwise transpose M) and put $\varepsilon_{ab} = (-1)^{M_{ab}}$. Summing over ordered pairs $a \neq a'$ and $b \neq b'$,

$$\sum \varepsilon_{ab} \varepsilon_{ab'} \varepsilon_{a'b} \varepsilon_{a'b'} = \sum_{a \neq a'} \left(\left(\sum_b \varepsilon_{ab} \varepsilon_{ab'} \right)^2 - |B| \right) \geq -|A|(|A| - 1)|B|.$$

There are $|A|(|A| - 1)|B|(|B| - 1)$ terms, so the average of $(-1)^{\text{minor}}$ is at least $-1/(|B| - 1)$, and $f \geq \frac{1}{2} - \frac{1}{2(|B| - 1)}$. $\square$

So $d(\Gamma) \geq 1 - 4s + 5s^2 - o(1) \geq 5/16 - o(1)$, with near-equality only for $s \approx 1/4$ and $f \approx 1/2$; these near-extremal members are Giraud's construction.

Theorem 4. (a) $t(5, 4) \geq \frac{35604499940047}{115448720916480} > 0.308400$, hence $\pi(K_5^{(4)}) < 0.691600$.

$$(b) \ t(6, 4) \geq \frac{9224492822869}{65970697666560} > 0.139827, \text{ hence } \pi(K_6^{(4)}) < 0.860173.$$

$$(c) \ t(7, 4) \geq \frac{7476698908057}{115448720916480} > 0.064762, \text{ hence } \pi(K_7^{(4)}) < 0.935238.$$

$$(d) \ t_2(5, 4) \geq \frac{48641756816487}{87960930222080} > 0.552992, \text{ hence } \pi(K_5^{4-}) < 0.447008.$$

$$(e) \ t_2(6, 4) \geq \frac{202684381843379}{923589767331840} > 0.219452, \text{ hence } \pi(K_6^{4-}) < 0.780548.$$

The denominators are $105 \cdot 2^{40}$ in (a) and (c), and $15 \cdot 2^{42}$, $5 \cdot 2^{44}$ and $105 \cdot 2^{43}$ in (b), (d) and (e); they come from the scaling of the integer certificates (Section 3.4).

Theorem 5. $\pi(K_6^{(5)}) < 0.767401$, $\pi(H_4^5) < 0.389709$ and $\pi(K_7^{(5)}) < 0.927918$. The exact rational bounds are given in Appendix A.

For $K_5^{(4)}$, Markström's bound is $1753/2380 < 0.736555$ [27]; repeated averaging from $T(18, 5, 4) \geq 807$ gives the unpublished 0.735614 , against which the gap removed is 91.5%. The entries "from [42]" in Table 1 use $T(n, p, 4) \geq \lceil nT(n-1, p, 4)/(n-4) \rceil$, starting from $T(16, 6, 4) \geq 190$ and $T(16, 7, 4) \geq 99$ in Sidorenko's tables and stopping at $n = 200\,000$. This

	this paper	previous upper bound	lower bound	gap removed
$\pi(K_5^{(4)})$	< 0.691600	$751/1020 < 0.736275$ [42]	$11/16$ [18]	91.6%
$\pi(K_6^{(4)})$	< 0.860173	< 0.894570 (from [42])	> 0.845898 [42]	70.7%
$\pi(K_7^{(4)})$	< 0.935238	< 0.944386 (from [42])	> 0.913476 [42]	29.6%
$\pi(K_5^{4-})$	< 0.447008	$1/2$ [28, 44]	> 0.428571 [44]	74.2%
$\pi(K_6^{4-})$	< 0.780548	< 0.852398 (derivable)	$11/16$ [18]	43.6%
$\pi(K_6^{(5)})$	< 0.767401	< 0.769678 [8]		
$\pi(H_4^5)$	< 0.389709	$2/5$ [28]		
$\pi(K_7^{(5)})$	< 0.927918	$14/15 < 0.933334$ [10]		

Table 1: The new bounds against the literature (upper bounds rounded up, lower bounds rounded down). The last column is the share of the previous interval [lower, upper] removed. Construction bounds for the three 5-graphs were not checked.

gives a valid bound because $T(n, p, 4)/\binom{n}{4}$ is non-decreasing; the same rule is used for K_6^{4-} and in Table 2. One step of de Caen’s averaging bound [11, Theorem 3] gives the weaker 0.895548 and 0.945575; his general bounds are $9/10$ and $19/20$ [10]. The constructions are [42, Ex. 3.4] for $K_6^{(4)}$ and $4677/5120$ [42, Prop. 6.1] for $K_7^{(4)}$. For $K_6^{(5)}$, the upper bound is $(43 + \sqrt{345})/80$, stated in [8, Theorem 3] with the proof omitted.

For K_6^{4-} , the derivable upper bound 0.852398 comes from repeated averaging $T_2(n, 6, 4) \geq \lceil nT_2(n-1, 6, 4)/(n-4) \rceil$, starting at $T_2(6, 6, 4) = 2$; we found no published bound specific to this graph. The lower bound holds because $K_5^{(4)} \subset K_6^{4-}$.

For K_5^{4-} the published bounds $3/7 \leq \pi(K_5^{4-}) \leq 1/2$ are those of Sidorenko [44], the upper one being the general bound $\pi(H_k^r) \leq (k-2)/r$ [28, Cor. 3.9]; they are restated in recent work [46]. The problem is the case $(n; 5, 4)$ of the Brown–Erdős–Sós problem for 4-graphs: a 4-graph is K_5^{4-} -free iff every five vertices span at most three edges. For K_6^{4-} (the case $(n; 6, 4)$ of the same problem) we found no published bound beyond the inclusions $K_5^{(4)} \subset K_6^{4-} \subset K_6^{(4)}$, and Sidorenko’s general bound $\pi(F) \leq 1 - 1/(e(F) - 1) = 12/13$ (as stated in [26]) is weaker than the averaging bound of Table 1.

The codegree-squared density. For a 4-graph G and a 3-set $T \subseteq V(G)$ let $\deg_G(T)$ be the number of edges containing T , and let $\text{co}_2(G) = \sum_T \deg_G(T)^2$, the sum over all 3-subsets of $V(G)$. Balogh, Clemen and Lidický [3] define the *codegree-squared density* of a 4-graph F as

$$\sigma(F) = \limsup_{n \rightarrow \infty} \max \left\{ \frac{\text{co}_2(G)}{\binom{n}{3}(n-3)^2} : G \text{ is an } F\text{-free 4-graph on } n \text{ vertices} \right\}.$$

They state $31/64 \leq \sigma(K_5^{(4)}) \leq 0.55241250$, the lower bound by Giraud’s construction and the upper bound by flag algebras, and conjecture that the lower bound is tight [3, Conjecture 8.2 of arXiv v3].

Theorem 6. $\sigma(K_5^{(4)}) = 31/64$.

The proof (Section 3.8) is a plain seven-vertex certificate for the objective $\kappa = 2d - \gamma$, a combination of densities on five vertices (quadratic in the codegrees). Unlike every other certificate in this paper, it gives a sharp bound, matched by Giraud’s construction; the minimum $33/64$ of $\kappa - \sum c_Q$ is attained exactly, and only, on the ten isomorphism classes of 7-vertex subgraphs of Giraud’s construction. That rigidity gives the following stability theorem.

Theorem 7. *For every $\delta > 0$ there are $\varepsilon > 0$ and n_0 , namely $\varepsilon = \delta/(2C_0)$ and $n_0 = \lceil 7 + 2(69C_0 + 24)/\delta \rceil$ with $C_0 < 7.81 \cdot 10^8$ as in Theorem 20, such that every $K_5^{(4)}$ -free 4-graph G on $n \geq n_0$ vertices with $\text{co}_2(G) \geq (31/64 - \varepsilon)\binom{n}{3}(n-3)^2$ becomes the complement of a Giraud system after changing at most $\delta\binom{n}{4}$ of its 4-sets. The Giraud system can be chosen with parts of sizes $n/2 \pm O(\sqrt{\varepsilon + 1/n})n$ and a fraction $1/2 + O(\varepsilon + 1/n)$ of even minors, that is, with a quasirandom matrix. (The roles of ε and δ are reversed with respect to Hypothesis 41.)*

Section 4 proves an explicit version (Theorem 20) from the margin of the certificate outside Giraud’s ten classes and from Theorem 16: a 4-graph on at least seven vertices is a Giraud system iff all its 7-vertex induced subgraphs are.

A catalogue of threshold families. The same method applies to every threshold family. Let $F_{p,\lambda}$ be the family of p -vertex 4-graphs with at least $\binom{p}{4} - \lambda + 1$ edges; a 4-graph is $F_{p,\lambda}$ -free iff its complement spans at least λ edges in every p -set, so $\pi(F_{p,\lambda}) = 1 - t_\lambda(p, 4)$. The cases $(5, 1)$, $(5, 2)$, $(6, 1)$, $(6, 2)$ and $(7, 1)$ are $K_5^{(4)}$, K_5^{4-} , $K_6^{(4)}$, K_6^{4-} and $K_7^{(4)}$.

Theorem 8. *The lower bounds on $t_\lambda(p, 4)$ in Table 2 hold; each is a plain seven-vertex certificate with exact rational entries, and the case $(5, 3)$ reproduces the known value $\pi(H_3^4) = 1/4$ [19] up to $1.7 \cdot 10^{-8}$.*

(p, λ)	minimal members of $F_{p,\lambda}$	$t_\lambda(p, 4) >$	previous	construction	gap removed
$(5, 3)$	H_3^4	0.749999983648	3/4 (exact)	3/4	validation
$(6, 3)$	$K_6^{(4)}$ minus 2 edges	0.290180099259	0.219452	5/16	76.0%
$(6, 4)$	$K_6^{(4)}$ minus 3 edges	0.356110768826	0.291249	0.410755	54.3%
$(6, 5)$	$K_6^{(4)}$ minus 4 edges	0.421440536123	0.358011	1/2	44.7%
$(6, 6)$	$K_6^{(4)}$ minus 5 edges	0.497081561103	0.409140	4/7	54.2%
$(6, 9)$	6-vertex, 7 edges	0.705293990330	0.609234	3/4	68.2%
$(6, 11)$	6-vertex, 5 edges	0.840815156178	0.752169	19/21	58.1%
$(7, 2)$	K_7^{4-}	0.101742874380	0.069349	0.154102	38.2%
$(7, 3)$	$K_7^{(4)}$ minus 2 edges	0.131341381781	0.093111	0.154102	62.7%
$(7, 4)$	$K_7^{(4)}$ minus 3 edges	0.161680760732	0.123665	7/27	28.0%

Table 2: Threshold families at $N = 7$ (lower bounds floored; $\pi(F_{p,\lambda}) = 1 - t_\lambda(p, 4)$). “Previous” is the best lower bound available without these runs (repeated averaging from small exact values, and our bounds for K_6^{4-} and $K_7^{(4)}$); “construction” is the best upper bound on $t_\lambda(p, 4)$ we know (see the text).

The upper bounds come from inclusions and simple constructions. If every 5-set spans at least μ edges, every 6-set spans at least 3μ (each 4-set lies in two of its six 5-subsets), and if every 6-set spans at least μ , every 7-set spans at least $\lceil 7\mu/3 \rceil$. So Giraud’s construction bounds $t_3(6, 4)$ by 5/16, Sidorenko’s constructions [42, 44] bound $t_2(7, 4)$, $t_3(7, 4)$ by 0.154102 and $t_6(6, 4)$ by 4/7, and the complement of the tournament 4-graph of Gunderson and Semeraro [19], in which every five vertices span 0 or 2 edges, bounds $t_9(6, 4)$ by 3/4. The remaining ones are elementary: the even bipartite 4-graph (the 4-sets meeting a fixed half of the vertices in 0, 2 or 4 points) has density 1/2 and every six vertices span at least five edges; three parts of relative sizes 0.18614, 0.40693, 0.40693 with all 4-sets inside a part and those of patterns $(3, 1, 0)$, $(3, 0, 1)$, $(1, 3, 0)$, $(1, 0, 3)$, $(2, 1, 1)$, $(0, 2, 2)$ give $t_4(6, 4) < 0.410755$; three equal parts with the 4-sets inside a part or with two points in each of two parts give $t_4(7, 4) \leq 7/27$; and the complement of the iterated complete 4-partite 4-graph gives $t_{11}(6, 4) \leq 19/21$. Apart from our own K_6^{4-} result we found nothing published on these families for $p = 6, 7$ and $\lambda \geq 2$, which

are the dense regime of the Brown–Erdős–Sós problem. Individually the bounds are minor; their value is as a systematic table.

The six-vertex barrier uses Definition 10 below; “plain” means that only sums of squares of flag densities are used, with no stationarity or other additional constraint.

Theorem 9. *No plain flag-algebra certificate on six vertices proves $t(5, 4) > 1/4$, $t(6, 4) > 1/10$, $t(7, 4) > 0$ or $t_2(5, 4) > 1/2$.*

In Turán form the six-vertex method cannot go below $3/4$, $9/10$ and $1/2$ for $K_5^{(4)}$, $K_6^{(4)}$ and K_5^{4-} , which are de Caen’s bounds and the general $(k - 2)/r$ bound, and it gives nothing for $K_7^{(4)}$. The values $1/4$, $1/10$ and $1/2$ are attained by six-vertex certificates, checked exactly by the first implementation and by an independent checker (Remark 30), so they are the exact six-vertex optima.

3 Flag-algebra certificates on seven vertices

3.1 The inequality

We use the flag-algebra inequality of Razborov [38] in an elementary form, written for 4-graphs and seven vertices; nothing below depends on the general theory.

A *type* of size s is a 4-graph σ on $[s]$. A σ -*flag* on $m \geq s$ vertices is a pair (F, θ) where F is a 4-graph on m vertices and $\theta: [s] \rightarrow V(F)$ is injective with $\theta(X) \in E(F) \Leftrightarrow X \in E(\sigma)$ for every 4-set $X \subseteq [s]$. Two σ -flags are isomorphic if a bijection of the vertex sets maps edges to edges and θ to θ' . Let $\mathcal{F}_\sigma^m$ be a finite list of pairwise non-isomorphic σ -flags on m vertices. For a 4-graph H , an injective $\theta: [s] \rightarrow V(H)$ and a set $U \subseteq V(H) \setminus \theta([s])$ of size $m - s$, write H_θ^U for the pair $(H[\theta([s]) \cup U], \theta)$; it is a σ -flag exactly when $H[\theta] = \sigma$, that is, when θ is an isomorphism from σ onto $H[\theta([s])]$.

Definition 10 (Plain certificate on N vertices). A *block* is a tuple $(s, m, \sigma, \mathcal{F}_\sigma^m, Q)$ with $2m - s \leq N$, a type σ of size s , a flag list $\mathcal{F}_\sigma^m$ and a positive semidefinite matrix Q indexed by $\mathcal{F}_\sigma^m$. For a 4-graph H on N vertices put

$$c_Q(H) = \mathbb{E}_{(\theta, U_1, U_2)} \left[Q(H_\theta^{U_1}, H_\theta^{U_2}) \right],$$

where (θ, U_1, U_2) is uniform among the injective maps $\theta: [s] \rightarrow V(H)$ and the ordered pairs of *disjoint* $(m - s)$ -subsets of $V(H) \setminus \theta([s])$, and where $Q(F, F')$ is the entry of Q if both arguments are σ -flags isomorphic to members of $\mathcal{F}_\sigma^m$, and 0 otherwise. A *plain certificate on N vertices* for (p, λ) is a finite list of blocks; its *value* is

$$b = \min_H \left(d(H) - \sum_{\text{blocks}} c_Q(H) \right),$$

the minimum over all (p, λ) -admissible 4-graphs H on N vertices.

The number of triples (θ, U_1, U_2) in an N -vertex graph is $\text{conf}(s, m) = \frac{N!}{(N-s)!} \binom{N-s}{m-s} \binom{N-m}{m-s}$.

Proposition 11. *If a plain certificate on N vertices for (p, λ) has value b , then $t_\lambda(p, 4) \geq b$.*

Proof. Let G be a (p, λ) -admissible 4-graph on $n \geq N$ vertices and fix one block. For an injective $\theta: [s] \rightarrow V(G)$ with $G[\theta] = \sigma$, let v_θ be the vector indexed by $\mathcal{F}_\sigma^m$ with $v_\theta(F) = \mathbb{P}_U[G_\theta^U \cong F]$, U a uniform $(m - s)$ -subset of $V(G) \setminus \theta([s])$; put $v_\theta = 0$ if $G[\theta] \neq \sigma$. Since $Q \succeq 0$, $v_\theta^\top Q v_\theta \geq 0$, and $v_\theta^\top Q v_\theta = \mathbb{E}_{U_1, U_2} Q(G_\theta^{U_1}, G_\theta^{U_2})$ with U_1, U_2 independent. Conditioning on $U_1 \cap U_2 = \emptyset$ changes this expectation by at most $2 \max |Q| \cdot \mathbb{P}[U_1 \cap U_2 \neq \emptyset] = O(1/n)$. Hence the average over a uniform configuration (θ, U_1, U_2) of G , with U_1, U_2 disjoint, is at least $-O(1/n)$.

A configuration occupies exactly $2m - s \leq N$ vertices. Every set of $2m - s$ vertices lies in the same number of N -subsets, and every N -subset contains $\text{conf}(s, m)$ configurations. So a uniform configuration of G is obtained by choosing a uniform N -subset S and then a uniform configuration of $G[S]$; the flag G_θ^U does not change when G is replaced by $G[S]$. Therefore $\mathbb{E}_S c_Q(G[S]) \geq -O(1/n)$ for each block. Also $d(G) = \mathbb{E}_S d(G[S])$. Each $G[S]$ is admissible, so

$$d(G) = \mathbb{E}_S \left(d(G[S]) - \sum c_Q(G[S]) \right) + \sum \mathbb{E}_S c_Q(G[S]) \geq b - O(1/n).$$

Applying this to extremal graphs and letting $n \rightarrow \infty$ gives $t_\lambda(p, 4) \geq b$. $\square$

No symmetry of Q is assumed, and flags or types missing from the lists only make the inequality weaker; the lists need not be complete for soundness (ours are complete; Section 3.6). Theorems 4 and 5 are Proposition 11 applied to the certificates of Appendix A, with the value b computed exactly as described below.

Remark 12. The proof uses the edge density only through the exact identity $d(G) = \mathbb{E}_S d(G[S])$ for a uniform N -subset S . So it applies verbatim to any objective ψ in place of d that is defined on all 4-graphs with at least N vertices and satisfies $\psi(G) = \mathbb{E}_S \psi(G[S])$ whenever $|V(G)| \geq N$: if the certificate with d replaced by ψ has value b , then $\psi(G) \geq b - O(1/n)$ for every (p, λ) -admissible G on $n \geq N$ vertices. Every induced density of a configuration on at most N vertices has this property, for instance the probability that a uniform sequence of $j \leq N$ distinct vertices spans a given labelled 4-graph, and so has every linear combination of such densities.

3.2 Blocks at seven vertices

At $N = 7$ the blocks with $m > s$ and $m \geq 4$ are $(s, m) \in \{(1, 4), (2, 4), (3, 4), (3, 5), (4, 5), (5, 6)\}$; the other blocks ($m \leq 3$ or $m = s$) have a single flag per type and only lower the bound. Table 3 gives the sizes. A type on $s \leq 3$ vertices has no edge, so these blocks have one type. A 4-graph on five vertices is determined up to isomorphism by its number of edges, so the types of the $(5, 6)$ block are indexed by $e(\sigma) \in \{0, \dots, 5\}$, restricted to the admissible ones. A σ -flag on $m = s + 1$ vertices is determined by the link of the free vertex, a set of triples of $[s]$; for $(5, 6)$ there are $2^{10} = 1024$ links. The $(5, 6)$ block needs $2m - s = 7$, so it does not exist at $N = 6$; in floating point it carries almost all of the gain (Section 7).

(s, m)	(1, 4)	(2, 4)	(3, 4)	(3, 5)	(4, 5)	(5, 6)
$\text{conf}(s, m)$	140	1260	2520	1260	5040	5040
$t(5, 4)$	2	2	2	23	15, 16	809, 854, 904, 960, 1024
$t(6, 4)$	2	2	2	24	16, 16	1023, 1024 ($\times 5$)
$t(7, 4)$	2	2	2	24	16, 16	1024 ($\times 6$)
$t_2(5, 4)$	2	2	2	19	11, 15	373, 463, 588, 768
$t_2(6, 4)$	2	2	2	24	16, 16	1013, 1023, 1024 ($\times 4$)

Table 3: Number of admissible flags per type at $N = 7$. For $(4, 5)$ the types are the empty and the one-edge 4-graph; for $(5, 6)$ they are the admissible 5-vertex types in increasing number of edges (1 to 5 edges for $t(5, 4)$, 0 to 5 for $t(6, 4)$, $t(7, 4)$ and $t_2(6, 4)$, 2 to 5 for $t_2(5, 4)$).

3.3 Finding a certificate

Only two facts about a certificate enter the proof: its matrices are positive semidefinite, and its value b is computed exactly (Sections 3.4 and 3.5). How the certificate was found does not matter for correctness. The best value of Definition 10 is a semidefinite program:

maximise b over positive semidefinite Q 's subject to $d(H) - \sum c_Q(H) \geq b$ for every admissible H . Writing $c_Q(H) = \langle Q, M_\sigma(H) \rangle$ with the flag-pair matrix $M_\sigma(H)$, there are up to 7013319 linear constraints and ten to twelve Gram blocks of order up to 1024. We do not solve this program directly. Following Jeong et al. [21, Section 4], we restrict each Q_σ to the cone generated by finitely many rank-one matrices $vv^\top$ (*cuts*), as in the column-generation scheme of Ahmadi, Dash and Hall [1], and the constraints to a working set W of graphs. We solve the dual of the restricted linear program, in the orientation of [21], with the dual simplex method of HiGHS [20]:

$$\min \sum_{H \in W} y_H d(H) \quad \text{s.t.} \quad \sum_{H \in W} y_H = 1, \quad \sum_{H \in W} y_H v^\top M_\sigma(H) v \geq 0 \text{ for every cut } v, \quad y \geq 0.$$

Because we name the objects after this program, our names are the mirror of those of [21]: our cuts are their added columns, and the graphs added to W are their cutting planes. The value b and the cut weights are the dual variables of this program. New cuts are eigenvectors of the most negative eigenvalues of the moment matrices $Y_\sigma = \sum_H y_H M_\sigma(H)$. New graphs are the most violated constraints, found by a scan of all admissible classes that never stores $M_\sigma(H)$ outside W . Every combination of cuts with nonnegative weights is positive semidefinite, so every full scan gives a valid floating-point bound, and we keep the best one. When the cuts stop improving the value, we freeze the cut set and add graphs until a full scan finds no violated constraint. We evaluate $c_Q(H)$ from unordered root sets with the automorphisms of σ , as in [21]; this is an identity for every Q and needs no symmetry of Q .

For $t(5, 4)$ the final working set has 4422 of the 3908438 admissible classes and 1337 cuts; Table 9 gives the floating-point value. The runs that give the certificates of Theorem 4(a) and (c) used at most two cores and 1.5 GB of memory and took between 67 and 75 minutes each, plus three minutes for the frozen-cut phase of $t(5, 4)$. The certificates of Theorem 4(b), (d) and (e) come from runs on 30-core cloud machines that resumed the saved states of earlier runs: for the final certificates, 150 minutes of cut generation for the three problems together, then the frozen-cut phase; about four machine-hours. For $t(6, 4)$ the certificate before the frozen-cut phase is the larger one, by $2.5 \cdot 10^{-8}$, and is the one we use. The code also supports the stationarity constraints of Razborov's differential method, but every certificate in this paper has them switched off (Section 7).

3.4 Exact rounding

Let $Q \approx \sum_i w_i u_i u_i^\top$ be the floating-point solution of a block in eigenform. We drop eigenvalues below a small relative threshold, round $a_i = \text{round}(M \sqrt{w_i} u_i)$ to an integer vector, and set $\tilde{Q} = A^\top A / M^2$, where the rows of A are the a_i and $M = 2^{20}$ or 2^{21} . Then $\tilde{Q}$ is rational and positive semidefinite by construction (compare [33]). The certificate is the list of integer matrices A ; the floating-point solution plays no further role. The value b is computed for $\tilde{Q}$ itself, so rounding may change the bound but never invalidates it. Rounding cost about $1.3 \cdot 10^{-7}$, $1.2 \cdot 10^{-7}$ and $2.0 \cdot 10^{-7}$ for $t(5, 4)$, $t(6, 4)$ and $t(7, 4)$, and $1.2 \cdot 10^{-7}$ and $9.3 \cdot 10^{-8}$ for $t_2(5, 4)$ and $t_2(6, 4)$ (Table 9).

3.5 Exact evaluation over all admissible graphs

All the numbers $\text{conf}(s, m)$ in Table 3 divide 5040, and $A^\top A$ has integer entries. So for every 7-vertex H the number $Z(H) = 5040 M^2 \sum c_{\tilde{Q}}(H)$ is an integer, and

$$d(H) - \sum c_{\tilde{Q}}(H) = \frac{144 M^2 e(H) - Z(H)}{5040 M^2}.$$

The value b is the minimum of this fraction over the admissible H , computed in integer arithmetic.

Lemma 13. *Let $R_1, \dots, R_q$ represent the isomorphism classes of (p, λ) -admissible 4-graphs on $\{0, \dots, 5\}$. Every (p, λ) -admissible 4-graph on seven vertices is isomorphic to a graph $R_i \cup L$ obtained by adding the vertex 6 with link L , a set of triples of $\{0, \dots, 5\}$.*

Proof. Delete any vertex v of H ; the rest is admissible, hence isomorphic to some R_i . Transport the isomorphism and send v to 6; the link of 6 is the image of the link of v . $\square$

Since $d - \sum c_{\tilde{Q}}$ is an isomorphism invariant, b is the minimum over the admissible graphs $R_i \cup L$, L ranging over all 2^{20} links. We computed b twice in the first implementation: over a list of canonical representatives of all admissible classes, produced by one-vertex extension with nauty canonical forms [29]; and over all admissible raw extensions $R_i \cup L$, which needs no canonical form. Both scans give the same minimum and the same maxima of Z on every group of graphs with equal edge count and equal value of a second invariant. Table 4 gives the sizes. The class counts were also confirmed by the labelled count $\sum_H 7!/|\text{Aut}(H)|$ and, in the second implementation, by Burnside’s lemma over S_7 .

	$t(5, 4)$	$t(6, 4)$	$t(7, 4)$	$t_2(5, 4)$	$t_2(6, 4)$
classes, 6 vertices	122	155	156	62	154
classes, 7 vertices	3 908 438	7 011 184	7 013 319	360 127	6 986 573
labelled, 7 vertices	19 199 206 747	34 352 419 335	$2^{35} - 1$	1 741 143 596	34 244 802 014
raw extensions $R_i \cup L$	86 952 880	162 481 445	163 577 855	10 058 620	160 859 017

Table 4: Sizes of the exhaustive scans (admissible graphs only). For $p = 7$ a 7-vertex graph is admissible unless it is empty; 7 013 320 is the number of 3-graphs on seven unlabelled vertices.

3.6 Independent verification

A second implementation, written separately and sharing no code with the first, re-checked every certificate. Its decisive check, the raw scan, reads only the certificate files (blocks, types, flag lists and the integer matrices A). It differs from the first implementation in every step that matters:

- its own edge order and its own encoding of the flags;
- its own 6-vertex representatives, by brute force over all 720 permutations of all 2^{15} labelled 4-graphs on six vertices, instead of nauty;
- an evaluator in C with 128-bit integers that enumerates every ordered injective θ with an exact labelled type test and every ordered disjoint pair (U_1, U_2) ; it does not use the automorphisms of σ ;
- a separate check that every flag decodes to a flag of its type, that the flags of a list are pairwise non-isomorphic and admissible, and that every list is complete.

For each of the five certificates it scanned all raw extensions of Table 4, found exactly the claimed minimum with the same group maxima as the first implementation, and evaluated the minimisers again with a pure-Python evaluator in exact fractions. Scanning the class lists of the first implementation gave the same result. Independently of the certificates, it recomputed the class counts by Burnside’s lemma. With $Q = 0$ it recovered the minimum edge counts 7, 3 and 1 for $p = 5, 6, 7$, the Turán numbers $T(7, p, 4)$.

As a sanity check of signs and normalisation, 100 000 random 7-subsets of a Giraud system with 150 points in each part gave a positive average of $\sum c_{\tilde{Q}}$, consistent with Proposition 11. This was done for $t(7, 4)$ and for earlier certificates of $t(5, 4)$, $t(6, 4)$ and $t_2(6, 4)$.

The second implementation's raw scans for $t_2(5, 4)$ and $t_2(6, 4)$ were run on both a cloud and a local machine, with identical totals, minimum, group maxima and checksum. For $t(6, 4)$ it ran on a cloud machine, using input files it had generated itself. Its C evaluator takes ten minutes to three and a half hours per raw scan on two cores, and about five minutes on 32 cores.

The ten catalogue certificates of Theorem 8 have the same form ($Q = A^\top A/M^2$ with $M = 2^{19}, \dots, 2^{22}$, the blocks of Section 3.2, no stationarity terms); their exact values are listed in Appendix A. For $(5, 3)$ the certificate is the one obtained before the frozen-cut phase (that phase produced no separate certificate there); for the other nine it is the frozen-cut one. The second implementation scanned the class lists and raw extensions of all ten on cloud machines. The decoded certificates and six-vertex representatives were bit-identical to its locally generated inputs. Raw scans for $(5, 3)$, $(6, 6)$, $(6, 9)$ and $(6, 11)$ were repeated locally, with identical results on every deterministic field. At four graphs per case (the two minimisers found and the next two values), the pure-Python evaluator and a third, independent evaluator agreed exactly.

3.7 Five-graphs

Definition 10 and Proposition 11 hold verbatim for r -graphs, with r -sets in place of 4-sets, and with the same proof. Theorem 5 uses them for 5-graphs on seven vertices, with all blocks satisfying $2m - s \leq 7$. A 5-graph on seven vertices corresponds to a graph on seven vertices, so there are only 1044 classes, and the semidefinite program was solved directly. The matrices Q were rounded through their eigendecomposition, eigenvalues and eigenvector entries to multiples of 2^{-40} , which keeps them positive semidefinite. The certificate files store the rational matrices Q themselves, so a verifier must test positive semidefiniteness; both implementations did so exactly, by an $LDL^\top$ factorisation over the rationals, block by block. Both implementations evaluated the certificates on all 2^{21} labelled 5-graphs on seven vertices. The analogous computations for H_3^5 and H_5^5 give only the general bounds $1/5$ and $3/5$, up to rounding.

3.8 The codegree-squared density

For a 4-graph G on $n \geq 5$ vertices let

$$\gamma(G) = \mathbb{P}[T \cup \{x\} \in E(G) \text{ and } T \cup \{y\} \in E(G)],$$

where T is a uniform 3-subset of $V(G)$ and (x, y) is a uniform ordered pair of distinct vertices outside T .

Lemma 14. *Let G be a 4-graph on $n \geq 5$ vertices and H its complement in $K_n^{(4)}$. Then*

$$\frac{\text{co}_2(G)}{\binom{n}{3}(n-3)^2} = \gamma(G) - \frac{\gamma(G) - d(G)}{n-3} \quad \text{and} \quad \gamma(G) = 1 - \kappa(H),$$

where $\kappa(H) = 2d(H) - \gamma(H)$.

Proof. Write $\deg_G(T)^2 = \deg_G(T)(\deg_G(T) - 1) + \deg_G(T)$. Summed over T , the first terms count the triples (T, x, y) with $x \neq y$ outside T and $T \cup \{x\}, T \cup \{y\} \in E(G)$, so they add up to $\binom{n}{3}(n-3)(n-4)\gamma(G)$; the second terms add up to $4e(G) = \binom{n}{3}(n-3)d(G)$. Dividing by $\binom{n}{3}(n-3)^2$ gives the first identity. The second is inclusion-exclusion: each of $T \cup \{x\}$ and $T \cup \{y\}$ is an edge of H with probability $d(H)$. $\square$

Both d and γ are densities of configurations on at most five vertices, so κ has the property of Remark 12 for $N = 7$. On seven vertices $\kappa(H) = (24e(H) - P(H))/420$, where $P(H) = \sum_T \deg_H(T)(\deg_H(T) - 1)$ and $420 = \binom{7}{3} \cdot 4 \cdot 3$ is the number of triples (T, x, y) .

Proof of Theorem 6. Upper bound. A 4-graph G is $K_5^{(4)}$ -free iff its complement H is a Turán $(n, 5, 4)$ -system. Below we describe a plain certificate on seven vertices for $(p, \lambda) = (5, 1)$ with d replaced by κ , of value exactly $b = 33/64$. By Remark 12, $\kappa(H) \geq 33/64 - O(1/n)$ for every Turán $(n, 5, 4)$ -system H , so by Lemma 14, since $\gamma(G) - d(G) \geq -1$,

$$\frac{\text{co}_2(G)}{\binom{n}{3}(n-3)^2} \leq \gamma(G) + \frac{1}{n-3} = 1 - \kappa(H) + \frac{1}{n-3} \leq \frac{31}{64} + O(1/n)$$

for every $K_5^{(4)}$ -free G on n vertices, and $\sigma(K_5^{(4)}) \leq 31/64$.

Lower bound. This is the construction of [3, Section 8.2]; we give an explicit instance. Let $n = 2^{k+1}$, split the vertices into halves A and B , let M be the 0/1 form of the Sylvester–Hadamard matrix of order 2^k , so that any two distinct rows of M differ in exactly half of the columns and any two distinct columns in exactly half of the rows, and let G be the complement of the Giraud system $\Gamma(A, B, M)$ (Definition 2); G is $K_5^{(4)}$ -free by Lemma 3. The edges of G are the 4-sets with three points in one part and one in the other, and the 4-sets $\{a, a', b, b'\}$ with an odd minor. Up to $O(1/n)$ a uniform 3-set T lies inside one part with probability $1/4$, and then $T \cup \{x\}$ and $T \cup \{y\}$ are both edges iff x and y lie in the other part, with probability $1/4$. Otherwise, say $T = \{a, a', b\}$, the set $T \cup \{x\}$ is an edge iff $x \in A$, or $x \in B$ and $M_{ax} + M_{a'x} \neq M_{ab} + M_{a'b}$ modulo 2; the second condition holds for exactly half of the columns x , whatever b is. So both are edges with probability $\frac{1}{4} + 2 \cdot \frac{1}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{4} = \frac{9}{16}$, and $\gamma(G) = \frac{1}{4} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{9}{16} + O(1/n) = \frac{31}{64} + O(1/n)$. By Lemma 14 the normalised $\text{co}_2(G)$ tends to $31/64$; since these G are $K_5^{(4)}$ -free, $\sigma(K_5^{(4)}) \geq 31/64$ (a subsequence suffices for the lim sup). $\square$

The certificate. Let ϕ be the law of the 7-vertex subgraph of Giraud’s construction on uniformly random vertices, with parts and matrix entries independent and uniform (Section 7): $\mathbb{E}_\phi \kappa = 33/64$ exactly (the computation above, carried out in this model, has no error term), and ϕ is supported on ten classes, all of them Turán systems. Every plain certificate has $b \leq \mathbb{E}_\phi(\kappa - \sum c_Q) \leq 33/64$, because $\mathbb{E}_\phi c_Q = \sum_\omega \mathbb{P}(\omega) v_\omega^\top Q v_\omega \geq 0$, the sum over the root configurations ω (the parts of the s root vertices and the matrix entries among them), where v_ω is the law of the rooted flag given ω (flags on disjoint vertex sets are independent given ω). A certificate of value $33/64$ must therefore be tight on all ten classes and satisfy $v_\omega^\top Q v_\omega = 0$, or equivalently $(Q + Q^\top)v_\omega = 0$, for every ω of positive probability. Our matrices are symmetric, so this condition is $Qv_\omega = 0$.

The generic rounding of Section 3.4 did not give a sharp certificate. A first certificate rounded that way, from a cloud run of Section 3.3 with this objective, has $b = 2267741359833/2^{42}$ and gives only $\sigma(K_5^{(4)}) < 31/64 + 3.2 \cdot 10^{-7}$. For every type τ the sharp certificate has the form

$$Q_\tau = W_\tau^\top W_\tau / 2^{72} + B_\tau^\top X_\tau B_\tau,$$

where the integer rows of W_τ are orthogonal to all vectors v_ω (the floating-point solution of a cutting-plane run restricted to such matrices, rounded with the pivot coordinates solved exactly), B_τ is an integer basis of the span of the flags occurring in Giraud’s construction intersected with the orthogonal complement of the v_ω , and $X_\tau = 2^{-20}I + \Delta_\tau$ with Δ_τ (entries below $4 \cdot 10^{-11}$) solved exactly so that $\kappa - \sum c_Q$ equals $33/64$ on each of the ten classes. The second term is present only for the three types of the $(5, 6)$ block with 1, 3 and 5 edges (dimensions 20, 5, 1), and each X_τ is strictly diagonally dominant, hence positive definite (also checked by an exact $LDL^\top$ factorisation); so $Q_\tau \geq 0$. The entries of all Q_τ have common denominator $57 \cdot 2^{152}$. Both implementations evaluated $\kappa - \sum c_Q$ exactly on all 3 908 438 admissible classes and on all 86 952 880 raw extensions of Table 4: the minimum is exactly

33/64, attained exactly on the ten classes of ϕ , and every other admissible 7-vertex graph exceeds it by at least $2.8 \cdot 10^{-5}$. Tightness on Giraud's construction is not used for validity; it only explains why the bound is attained.

4 Stability for the codegree-squared density

We prove Theorem 7. Except in Theorem 16 and its lemmas, H is a Turán $(n, 5, 4)$ -system, the complement of a $K_5^{(4)}$ -free G , and κ and γ are as in Lemma 14. On five vertices, $\kappa = g(e)$ with $g(e) = e(9 - e)/20$, because two distinct 4-subsets of a 5-set meet in a 3-set; so $\kappa(H) = \mathbb{E}_S g(e(H[S]))$ over uniform 5-sets S , and $g(1), \dots, g(5) = \frac{2}{5}, \frac{7}{10}, \frac{9}{10}, 1, 1$. Giraud systems (Definition 2) are *odd*: every five vertices span an odd number of edges (Lemma 3); and induced subgraphs of Giraud systems are Giraud systems. We call H *locally Giraud* if every 7-vertex induced subgraph is a Giraud system.

Finite facts. The following were checked exhaustively by two independent programs, which both enumerate Giraud systems from Definition 2; to extend by a vertex, the first solves the parity system for the new link and tests every 7-set, the second joins the admissible link patterns over all 6-subsets.

- (G7) There are 2404 labelled Giraud systems on seven vertices, in 10 isomorphism classes; these are exactly the ten classes on which the certificate of Theorem 6 is tight. Only one class has more than one partition: the 4-graph whose edges are the complements of the seven lines of a Fano plane, with seven partitions of sizes (3, 4).
- (G8) There are 32981 labelled Giraud systems on eight vertices, in 22 classes. Only one class has more than one partition: the Steiner quadruple system $S(3, 4, 8)$ (the planes of $AG(3, 2)$), with seven partitions of sizes (4, 4); every 5-set of it spans exactly one edge.
- (Cn) For $n = 8, 9, 10$, every locally Giraud 4-graph on n vertices is a Giraud system. (For $n = 10$ the second program tested all extensions of all 604426 labelled Giraud systems on nine vertices and found exactly the 15636107 labelled Giraud systems on ten vertices.)

Proposition 15. *Let $\mu_0 = 176165518826891630107/6280747422216628134215680 > 2.8048 \cdot 10^{-5}$, the least value of $\kappa - \sum c_Q - 33/64$ over the admissible 7-vertex graphs that are not Giraud systems, for the certificate of Theorem 6. For every Turán $(n, 5, 4)$ -system H on $n \geq 10$ vertices,*

$$\mu_0 q(H) \leq \xi(H), \quad \text{where } \xi(H) = \kappa(H) - \frac{33}{64} + \frac{68}{n-7}$$

and $q(H)$ is the fraction of 7-sets S such that $H[S]$ is not a Giraud system.

Proof. By Section 3.8, $\kappa - \sum c_Q \geq 33/64$ on every admissible 7-vertex graph, with equality exactly on ten classes, which by (G7) are the Giraud systems; so it is $\geq 33/64 + \mu_0$ on the others; μ_0 was computed by an exact scan of all 3908438 classes and agrees with the independent evaluator of Section 3.8. Since $\kappa(H) = \mathbb{E}_S \kappa(H[S])$ over 7-sets, $\kappa(H) \geq 33/64 + \mu_0 q(H) + \sum_{\text{blocks}} \mathbb{E}_S c_Q(H[S])$. For one block, $\mathbb{E}_S c_Q(H[S]) = \langle Q, \bar{M} \rangle$, where $\bar{M}$ is the joint law of the two flags at a uniform root θ with independent uniform U_1, U_2 conditioned to be disjoint. If p is the probability that U_1 and U_2 meet, then, with v_θ as in the proof of Proposition 11, $\mathbb{E}_\theta[v_\theta v_\theta^\top] = (1 - p)\bar{M} + pX$ with $X \geq 0$ entrywise of total mass at most 1, so $\langle Q, \bar{M} \rangle \geq -\frac{p}{1-p} \max_a Q_{aa}$, since $Q \succeq 0$. The largest diagonal entries of the matrices Q , summed over the types of each block, are 61.2023, 0.9435, 1.4008 and 0.0003 for (5, 6), (4, 5), (3, 5) and (1, 4) (exact values from the certificate; the blocks (2, 4) and (3, 4) are zero), and $p/(1 - p)$ is $1/(n - 6)$, $1/(n - 5)$, $(4n - 18)/((n - 5)(n - 6)) \leq 4/(n - 7)$ and

at most $12/(n-7)$ for these blocks when $n \geq 10$. The total is at most $67.76/(n-7)$, so $\kappa(H) \geq 33/64 + \mu_0 q(H) - 68/(n-7)$. $\square$

The margin μ_0 is attained by an odd 7-vertex graph with eleven edges that is not a Giraud system; it is the bottleneck of all constants below.

4.1 Giraud systems are determined by their seven-vertex subgraphs

Theorem 16. *For $n \geq 7$, a 4-graph on n vertices is a Giraud system if and only if it is locally Giraud.*

For a 4-graph H and a vertex v , the *link* L_v is the 3-graph $\{R : R \cup \{v\} \in E(H)\}$ on $U = V(H) \setminus \{v\}$. Say that a 3-graph L on U has *form F with representation $A' \subseteq U$* , and $B = U \setminus A'$, if (F1) every triple inside A' is in L , (F2) no triple with exactly two points in A' is in L , (F3) no triple inside B is in L , and (F4) for $a \in A'$ and distinct $b_1, b_2, b_3 \in B$, an odd number of the triples $ab_i b_j$ are in L . By (F4) the graph $\{bb' : abb' \in L\}$ on B has one or three edges on every three vertices, so it is a disjoint union of at most two cliques.

Lemma 17. *If $H = \Gamma(A, B, M)$ and $v \in A$, then L_v has form F with representation $A \setminus \{v\}$.*

Proof. A triple inside $A \setminus \{v\}$ completes a 4-set inside A ; a triple with two points in A and one in B , or inside B , completes a $(3, 1)$ or a $(1, 3)$ set; and $\{v, a, b, b'\}$ is an edge iff $M_{vb} + M_{ab} = M_{vb'} + M_{ab'}$ modulo 2, an equivalence relation on B with at most two classes, which gives (F4). $\square$

Lemma 18. *If $A'_1 \neq A'_2$ are two representations of the same 3-graph L , then $|A'_1| \leq 3$ and $|A'_2| \leq 3$.*

Proof. By (F1)–(F3) every triple of L meets each A'_i in one or three points. Let $X = A'_1 \cap A'_2$, $Y = A'_1 \setminus A'_2$, $Z = A'_2 \setminus A'_1$; $Y \cup Z \neq \emptyset$. The triples inside A'_1 are in L , so none of them meets A'_2 in zero or two points; hence $|Y| \leq 2$, and $|X| \leq 1$ unless $Y = \emptyset$. Symmetrically $|Z| \leq 2$, and $|X| \leq 1$ unless $Z = \emptyset$. So $|X| \leq 1$ and $|A'_i| \leq 3$. $\square$

Lemma 19. *If $\Gamma(A, B, M) = \Gamma(A_2, B_2, M_2)$ and $|A| \geq 5$, then $\{A, B\} = \{A_2, B_2\}$. In particular a Giraud system on $n \geq 9$ vertices has a unique partition.*

Proof. Take $v \in A$ and let P_2 be the part of the second partition containing v . By Lemma 17, applied to both representations (with the parts swapped if $v \in B_2$), $A \setminus \{v\}$ and $P_2 \setminus \{v\}$ both represent L_v . The first has at least four points, so by Lemma 18 they are equal, and $P_2 = A$. $\square$

Proof of Theorem 16. Induced subgraphs of Giraud systems are Giraud systems, which gives one direction. For the other, $n = 7$ is the definition and $n = 8, 9$ are (C8) and (C9). Restrictions of a representation $\Gamma(A, B, M)$ to a subset W are representations of $H[W]$. Let H be locally Giraud on $n \geq 10$ vertices. We first show that there is a partition Π of V such that, for every vertex w , $H - w$ is a Giraud system whose unique partition π_w is the restriction of Π .

For $n \geq 11$ each $H - w$ is a Giraud system by induction, with a unique partition π_w (Lemma 19). For $w \neq w'$ the restrictions of π_w and $\pi_{w'}$ to $V \setminus \{w, w'\}$ ($n - 2 \geq 9$ vertices) are partitions of the same Giraud system, hence equal. So “ $x \sim y$ iff x and y lie in the same part of π_w for some $w \notin \{x, y\}$ ” does not depend on w ; any three points lie in a common $V \setminus \{w\}$, so $\sim$ is an equivalence relation with at most two classes, and these form Π .

For $n = 10$ each $H - w$ is a Giraud system by (C9), with a unique partition π_w of sizes $(a, 9-a)$. *Case 1: some $\pi_{w_0} = \{P, Q\}$ has $|P| \geq 6$.* For $x \neq w_0$, the restriction $\{P \setminus \{x\}, Q \setminus \{x\}\}$ of π_{w_0} to $V \setminus \{x, w_0\}$ has a part of size at least five, so it is the only partition there, and π_x is

obtained from it by adding w_0 to one side, say to the side of P or of Q . If w_0 joins P in π_x , then for $x' \neq x, w_0$ the restriction of π_x to $V \setminus \{x, x'\}$ has the part $(P \setminus \{x, x'\}) \cup \{w_0\}$ of size at least five, so it is the unique partition there, and w_0 joins P in $\pi_{x'}$ too. So the side of w_0 is the same for all x , and $\Pi = \{P \cup \{w_0\}, Q\}$ or $\{P, Q \cup \{w_0\}\}$. *Case 2: every π_w has sizes $(5, 4)$.* Fix $w_0, \pi_{w_0} = \{P, Q\}$ with $|P| = 5$, and $R = Q \cup \{w_0\}$. For $y \in Q$, the restriction $\{P, Q \setminus \{y\}\}$ to $V \setminus \{w_0, y\}$ is the unique partition there, so π_y is $\{P, R \setminus \{y\}\}$ or $\{P \cup \{w_0\}, Q \setminus \{y\}\}$, and the latter has sizes $(6, 3)$; so $\pi_y = \{P, R \setminus \{y\}\}$ for every $y \in R$. Now let $x \in P$. For $y \in R$ the 8-set $V \setminus \{x, y\}$ has the partition $\{P \setminus \{x\}, R \setminus \{y\}\}$; call y good if this is its only partition. If y is good, π_x is $\{P \setminus \{x\}, R\}$ or $\{(P \setminus \{x\}) \cup \{y\}, R \setminus \{y\}\}$, so two good y force $\pi_x = \{P \setminus \{x\}, R\}$. Otherwise at least four $y \in R$ are bad, and by (G8) every 5-subset of $V \setminus \{x, y\}$ spans exactly one edge for each bad y . The 5-part P_x of π_x spans five edges, so it contains every bad y . If all five are bad, $P_x = R$. If exactly four are bad, forming Y , then $P_x = Y \cup \{z\}$; if $z \in R$ then $P_x = R$, and $z \in P \setminus \{x\}$ is impossible: for the good y_5 the restriction $\{Y \cup \{z\}, P \setminus \{x, z\}\}$ of π_x to $V \setminus \{x, y_5\}$ has a 5-part, so it is the unique partition there, but the restriction of π_{y_5} is $\{P \setminus \{x\}, Y\}$. So $\pi_x = \{P \setminus \{x\}, R\}$ in all cases, and $\Pi = \{P, R\}$.

Gluing. Let $\Pi = \{A, B\}$. If $B = \emptyset$, every $H - w$ is complete, and so is H . Otherwise fix $a_0 \in A, b_0 \in B$, and put $M_{ab} = 1 - [\{a_0, a, b_0, b\} \in E(H)]$ for $a \neq a_0, b \neq b_0$, and $M = 0$ on row a_0 and column b_0 . For a 4-set T choose $w \notin T \cup \{a_0, b_0\}$. Then $H - w = \Gamma(A \setminus \{w\}, B \setminus \{w\}, M_w)$, and after switching M_w to vanish on row a_0 and column b_0 its entries are those of M (they are read off the same edges $\{a_0, a, b_0, b\}$). So $T \in E(H)$ iff $T \in E(\Gamma(A, B, M))$, and $H = \Gamma(A, B, M)$. $\square$

The case $n = 10$, the only one used below, was also checked exhaustively by both programs (C10). On at least seven vertices, the co-Fano class and $S(3, 4, 8)$ are the only Giraud systems with more than one partition (by (G7), (G8) and Lemma 19), which is why $n = 8, 9$ are done by computer and $n = 10$ needs Case 2. Chen, Hou, Liu, Zhang and Zhao [7, Lemma 3.3] prove a six-vertex local-to-global lemma of the same kind for complete odd-bipartite 4-graphs.

4.2 Robust stability

Theorem 20. *For every Turán $(n, 5, 4)$ -system H on $n \geq 10$ vertices there is a Giraud system Γ on $V(H)$ with*

$$|E(H) \triangle E(\Gamma)| \leq \left(C_0 \xi(H) + \frac{24}{n}\right) \binom{n}{4}, \quad C_0 = \frac{21894}{\mu_0} < 7.81 \cdot 10^8.$$

Consequently every $K_5^{(4)}$ -free G on $n \geq 10$ vertices with $\text{co}_2(G) \geq (31/64 - \varepsilon) \binom{n}{3} (n-3)^2$ is within $(C_0(\varepsilon + \frac{1}{n-3} + \frac{68}{n-7}) + \frac{24}{n}) \binom{n}{4}$ changed 4-sets of the complement of a Giraud system.

Proof. A frame is an ordered 6-tuple $\Phi = (a_0, a_1, a_2, a_3, a_4, b_0)$ of distinct vertices such that $\{a_0, \dots, a_4\}$ spans five edges and $\{a_0, a_1, a_2, b_0\}$ is not an edge. It defines a Giraud system $\Gamma_\Phi = \Gamma(A_\Phi, B_\Phi, M_\Phi)$: A_Φ consists of a_0, a_1, a_2 and the vertices x with $\{a_0, a_1, a_2, x\} \in E(H)$, B_Φ of the others (so $a_3, a_4 \in A_\Phi$ and $b_0 \in B_\Phi$), M_Φ vanishes on row a_0 and column b_0 , and $M_\Phi(a, b) = 1 - [\{a_0, a, b_0, b\} \in E(H)]$ otherwise.

Claim: if T is a 4-set disjoint from Φ and $H[\Phi \cup T]$ is a Giraud system, then $T \in E(H)$ iff $T \in E(\Gamma_\Phi)$. Write $H[\Phi \cup T] = \Gamma(A', B', M')$. A 5-set spans five edges only inside a part, so $\{a_0, \dots, a_4\} \subseteq A'$, say, and $b_0 \in B'$ because $\{a_0, a_1, a_2, b_0\}$ is not an edge. For $x \in T$, $\{a_0, a_1, a_2, x\}$ is an edge iff $x \in A'$, so $A' = A_\Phi \cap (\Phi \cup T)$. After switching M' to vanish on row a_0 and column b_0 , the minor of $\{a_0, a, b_0, b\}$ is $M'(a, b)$, so $M' = M_\Phi$ on $\Phi \cup T$, and $H[\Phi \cup T] = \Gamma_\Phi[\Phi \cup T]$.

Counting. Call a 10-set bad if it does not induce a Giraud system; by Theorem 16 (at $n = 10$) it then contains a 7-set that does not induce one, so there are at most $\binom{n}{7} \binom{n-7}{3} q(H) =$

$120q(H)\binom{n}{10}$ bad 10-sets. A pair of a frame Φ and a 4-set T disjoint from it with $[T \in E(H)] \neq [T \in E(\Gamma_\Phi)]$ makes $\Phi \cup T$ bad, and a 10-set arises from at most $(10)_6$ such pairs. Let ϱ be the number of frames divided by $(n)_6$. Averaging over frames, some frame Φ has at most $120q(H)\binom{n}{10}(10)_6/(\varrho(n)_6) = (120q(H)/\varrho)\binom{n-6}{4}$ such 4-sets T , and at most $6\binom{n-1}{3} = \frac{24}{n}\binom{n}{4}$ further 4-sets meet Φ . So $|E(H) \triangle E(\Gamma_\Phi)| \leq (120q(H)/\varrho + 24/n)\binom{n}{4}$.

Frame density. Frames live on six vertices, so ϱ is the average of the frame densities of the 10-vertex induced subgraphs. A Giraud system on ten vertices that is not complete has parts of sizes $a \geq 10 - a \geq 1$ and at least $(a)_5(10 - a) + (10 - a)_5 a \geq 1200 = (10)_6/126$ frames. Complete 10-sets have $\kappa = 1$ and every admissible graph has $\kappa \geq 2/5$, so the fraction w of complete 10-sets satisfies $\kappa(H) \geq w + \frac{2}{5}(1 - w)$. Hence $\varrho \geq (1 - \frac{5}{3}(\kappa(H) - \frac{2}{5}) - 120q(H))/126$.

Constants. If $\xi(H) \leq \mu_0/1200$, then $120q(H) \leq 1/10$ by Proposition 15 and $\kappa(H) \leq 33/64 + 1/100$, so $\varrho \geq \frac{221}{320} \cdot \frac{1}{126}$ and $120q(H)/\varrho \leq 21893.3 q(H) \leq C_0 \xi(H)$. If $\xi(H) > \mu_0/1200$, the bound exceeds $\binom{n}{4}$. For the second statement, Lemma 14 and $d - \gamma \leq 1$ give $\kappa(H) - 33/64 = 31/64 - \gamma(G) \leq 31/64 - \text{co}_2(G)/(\binom{n}{3}(n - 3)^2) + 1/(n - 3)$. $\square$

Proof of Theorem 7. By Theorem 20 it suffices that $C_0\varepsilon \leq \delta/2$ and $(69C_0 + 24)/(n - 7) \leq \delta/2$. For the structure, let $\Gamma = \Gamma(A, B, M)$ be the Giraud system of Theorem 20, $s = |A||B|/n^2$ and f its fraction of even minors. Counting the 5-sets by type, as in Lemma 3, gives

$$\kappa(\Gamma) = 1 - 3s + 3s^2 + \frac{5}{2}(3f - 1)s^2 + O(1/n), \quad \kappa(\Gamma) - \frac{33}{64} \geq \frac{7}{8}(\frac{1}{4} - s) + \frac{15}{2}s^2(f - \frac{1}{2}) - O(1/n),$$

and $f \geq \frac{1}{2} - O(1/n)$ when both parts have at least two points (Lemma 3); otherwise $s = O(1/n)$. Changing one 4-set changes κ by at most $2/\binom{n}{4}$, because $|g(e) - g(e \pm 1)| \leq 2/5$. So $\kappa(\Gamma) - 33/64 = O(\varepsilon + 1/n)$, hence $\frac{1}{4} - s = (\frac{1}{2} - |A|/n)^2 = O(\varepsilon + 1/n)$ and $f = \frac{1}{2} + O(\varepsilon + 1/n)$. $\square$

The condition $f = \frac{1}{2} + o(1)$ is the 4-cycle condition: the ± 1 matrix $((-1)^{M_{ab}})$ has $\sum_{a \neq a'} \langle r_a, r_{a'} \rangle^2 = o(|A|^2|B|^2)$ for its rows r_a , the 4-cycle form of quasirandomness [9], invariant under switching. So near-extremal $K_5^{(4)}$ -free graphs are close to the complement of Giraud's construction: balanced parts and a quasirandom matrix. In particular $d(G) = 11/16 + O(\varepsilon + 1/n)$, by the density formula of Lemma 3. A uniformly random matrix gives some balanced Giraud system with $f \leq \frac{1}{2}$, so every $K_5^{(4)}$ -free G with the largest $\text{co}_2(G)$ on n vertices has $\varepsilon = O(1/n)$, and by Theorem 20 it is within $O(n^3)$ changed 4-sets of the complement of a Giraud system.

Remark 21. The constants are explicit but not optimised, and they are weak because μ_0 is small: Proposition 15 says something only for $n > 2.4 \cdot 10^6$, and Theorem 20 only for $n > 5.38 \cdot 10^{10}$ and $\varepsilon < 1.3 \cdot 10^{-9}$. The qualitative statement, Theorem 7, holds with ε linear in δ , and its proof uses no removal lemma. A certificate with a larger margin outside Giraud's ten classes would improve all constants linearly.

Remark 22. The exact structure for large n remains open: we expect that every $K_5^{(4)}$ -free 4-graph with the largest codegree-squared sum is the complement of a Giraud system. The perfect-stability method of Pikhurko, Sliachan and Tyros [35] does not apply as it stands. It is written for graphs and needs a finite pattern whose blow-ups attain the optimum, while the near-extremal graphs here involve a quasirandom matrix (we expect that no blow-up of a finite pattern attains $31/64$, but we have not proved it). Quasirandom extremal structure also occurs for the codegree threshold of K_4^{3-} , the 3-graph on four vertices with three edges, [14], with stability towards the 3-graph of a quasirandom tournament, and Bodnár et al. [5] establish perfect stability for the inducibility of six-vertex graphs in all solved cases whose extremal construction has no quasirandom parts. Moreover, the natural analogue of the flip-aversion condition of [35] fails: in a balanced Giraud system whose matrix is quasirandom apart from two equal rows a, a' and two equal columns b, b' (so $\kappa = 33/64 + O(1/n)$), deleting the cross edge $\{a, a', b, b'\}$ keeps every 5-set spanning an edge, gives a graph that is not odd, hence not

a Giraud system, and lowers κ by exactly $(n-4)/(5\binom{n}{5})$ (for one matrix on 14 vertices, from 1061/2002 to 1059/2002). Finally, Theorems 7 and 20 concern κ , not the edge density: they do not imply Hypothesis 41. By the Cauchy–Schwarz inequality $\gamma(H) \geq d(H)^2 - O(1/n)$, so Theorem 6 gives only $t(5, 4) \geq 1 - \sqrt{31}/8 > 0.30402$, weaker than Theorem 4(a).

5 Further codegree-squared densities

For a 3-graph F the codegree-squared density $\sigma(F)$ is defined as in Section 2, with pairs in place of triples and the normalisation $\binom{n}{2}(n-2)^2$ [3, eq. (1) of arXiv v3]. The proof of Lemma 14 gives, for a 3-graph G on $n \geq 4$ vertices, $\text{co}_2(G)/(\binom{n}{2}(n-2)^2) = \gamma(G) - (\gamma(G) - d(G))/(n-2)$, where $\gamma(G)$ is now the probability that $P \cup \{x\}$ and $P \cup \{y\}$ are both edges for a uniform pair P and a uniform ordered pair (x, y) of distinct vertices outside P . Let J_t be the 3-graph on $t+1$ vertices consisting of a vertex u and all triples $\{u, a, b\}$ with a, b among the other t vertices; a 3-graph is J_t -free iff no vertex link contains K_t . Let $K_5^<$ be the 3-graph on five vertices with eight edges whose two missing edges meet in exactly two vertices [3, Section 3.20 of arXiv v3], let $K_5^=$ and K_5^{3-} (K_5^- in [3]) be obtained from $K_5^{(3)}$ by deleting two edges that meet in exactly one vertex, and one edge, respectively [3, Sections 3.19 and 3.21 of arXiv v3] (as $K_5^= \subset K_5^{3-}$, $\sigma(K_5^=) \leq \sigma(K_5^{3-})$), and let C_5 be the tight 5-cycle, with edges $\{i, i+1, i+2\}$ modulo 5 [3, Table 2 of arXiv v3]. Balogh, Clemen and Lidický conjecture $\sigma(J_4) = 0.28$ [3, Conjecture 3.4 of arXiv v3], on the basis of the iterated blow-up of the complement of the Fano plane of Bollobás, Leader and Malvenuto [6], and $\sigma(K_5^<) = 1/3$ [3, Conjecture 3.7 of arXiv v3], the value of Turán’s construction for the tetrahedron $K_4^{(3)} \subset K_5^<$.

Proposition 23. $\sigma(J_4) \geq 16/57 = 0.2807017\dots$, $\sigma(J_5) \geq 36/91 = 0.3956043\dots$ and $\pi(J_5) \geq 3/5$.

Proof. For J_4 take the iterated blow-up of the complement of the Fano plane (iterated inside every part; density $1/2$); for J_5 the iterated blow-up of the complement of the affine plane $AG(2, 3)$. In both, every vertex link is properly $(t-1)$ -colourable, so it contains no K_t : write vertices as words over the points of the plane and colour a vertex y of the link of x by the line through x_i and y_i , where i is the first coordinate in which x and y differ. (Iterated blow-ups of J_t -free non-degenerate 3-graphs are J_t -free in general [15].) For the first construction on $n = 7^L$ vertices, with $m = 7^{L-1}$, the codegree-squared sum satisfies $c_L = 7c_{L-1} + 336m^4$, since each of the 21 pairs of distinct parts contributes m^2 vertex pairs of codegree $4m$ (four of the five other points of the Fano plane complete the pair to a non-line), and pairs inside a part have all their codegree inside the part; so $\text{co}_2 = \frac{8}{57}(n^4 - n)$ and the normalised value tends to $16/57$; the second gives $36/91$ and edge density $3/5$. The best previous lower bounds we found are $\pi(J_5) \geq 1/2$ and $\sigma(J_5) \geq 0.28$ [3, Section 3.16 of arXiv v3], against the upper bounds 0.64474184 and 0.4427457 of [3]. $\square$

Remark 24. The value 0.28 in [3, Conjecture 3.4 of arXiv v3] comes from an arithmetic slip, identical in all three arXiv versions: the sum $\sum_i (1/7^4)^i$ there should be $\sum_i (1/343)^i$ for the construction described (their sum is that of the variant iterated inside a single part). So the natural corrected conjecture is $\sigma(J_4) = 16/57$. The same slip affects the bound $\sigma(J_5) \geq 0.28$ of [3]. The construction for J_5 in Proposition 23 is the next case of the same construction; we found it nowhere in the literature, but it is the obvious analogue and may be folklore.

Theorem 25. *The upper bounds in the last column of Table 5 hold; their exact values are the*

three below and those in Appendix A. In particular

$$\begin{aligned}\sigma(J_4) &\leq \frac{9061620987350745659}{32281802128991715328} < 0.280703690307, \\ \sigma(K_5^{\leq}) &\leq \frac{1375676321453}{4123168604160} < 0.333646, \\ \sigma(K_5^{4-}) &\leq \frac{6281420455337}{30786325577728} < 0.204033.\end{aligned}$$

F	lower bound	previous upper bound	this paper
J_4	$16/57 > 0.280701$ (Prop. 23)	0.28079696 [3]	0.2807037
$K_5^{\leq}$	$1/3$ [3]	0.3402107 [3]	0.3336455
$K_5^{\equiv}$	$2909/8127 > 0.357942$ [3]	0.388725 [3]	0.3844257
K_5^{3-}	$2909/8127 > 0.357942$ [3]	0.41961743 [3]	0.4059302
C_5	> 0.251940 [3]	0.25310457 [3]	0.2523050
$K_6^{(3)}$	> 0.734799 [3]	0.7535963 [3]	0.7423183
J_5	$36/91 > 0.395604$ (Prop. 23)	0.4427457 [3]	—
K_5^{4-}	$1/5$ (construction of [44])	$1/4$ (derivable)	0.2040329
K_6^{4-}	> 0.501220 (Rem. 27)	< 0.852398 (via π)	0.6120380
$K_6^{(4)}$	> 0.719219 (Rem. 27)	< 0.894570 (via π)	0.7437531
$K_7^{(4)}$	> 0.834439 (via π)	< 0.944386 (via π)	0.8771552

Table 5: Codegree-squared densities: upper bounds from plain certificates on seven vertices (rounded up) against the literature (upper bounds of [3] as printed there, other upper bounds rounded up, lower bounds rounded down). For the 4-graphs, “via π ” uses $\pi(F)^2 \leq \sigma(F) \leq \pi(F)$ (Cauchy–Schwarz, and $\deg_G(T) \leq n-3$) with the previous bounds of Table 1; with the bounds of Theorem 4 the upper ones become 0.780548, 0.860173 and 0.935238. The lower bound for $K_7^{(4)}$ is $(4677/5120)^2$, from [42, Prop. 6.1]. For C_5 the construction of [3, Section 3.17 of arXiv v3] has the edges xyz with $x, y \in B$, $z \in A$, $|B| = bn$, iterated inside A (the part sizes and the part of the iteration are interchanged there; the value is right). For J_5 our seven-vertex run was stopped before it reached the bound 0.4427457 of [3]; we claim no bound.

Proof. Definition 10, Proposition 11 and Remark 12 hold verbatim for 3-graphs and for F -free graphs in place of (p, λ) -admissible ones: the proof uses only that the class of graphs is closed under induced subgraphs. For the 3-graphs the blocks are the nine pairs (s, m) with $s < m$, $m \geq 3$ and $2m - s \leq 7$, and the objective is $1 - \gamma$, a density on four vertices; a certificate of value b gives $\gamma(G) \leq 1 - b + O(1/n)$ for every F -free G , hence $\sigma(F) \leq 1 - b$ by the identity above. For the 4-graphs, a 4-graph is K_5^{4-} -, K_6^{4-} -, $K_6^{(4)}$ - or $K_7^{(4)}$ -free iff its complement is $(5, 2)$ -, $(6, 2)$ -, $(6, 1)$ - or $(7, 1)$ -admissible, and the certificate is one for that (p, λ) with d replaced by κ , as in the proof of Theorem 6. Each certificate has $Q = A^\top A/M^2$ (Appendix A), and the stated fractions are $1 - b$. An independent implementation rebuilt every Q from the certificate files and computed the exact minimum b by a raw scan of all one-vertex extensions of its own 6-vertex representatives (29 092 719, 40 527 893 and 10 058 620 admissible extensions for J_4 , $K_5^{\leq}$ and K_5^{4-} , and from 5 635 379 for C_5 to 163 577 855 for $K_7^{(4)}$ for the others), and again over the class lists (except for K_6^{4-}), with the same group maxima. $\square$

The lower bound $1/5$ for $\sigma(K_5^{4-})$ in Table 5 comes from Sidorenko’s doubling construction [44]: with $h = n/2$, its codegree-squared sums satisfy $c_{L+1} = 2c_L + 2\binom{h}{2}h(h-1)^2$, so $\text{co}_2 \sim n^5/30$. For K_5^{4-} we found no published upper bound for σ . The bound $1/4$ is easy: if G is K_5^{4-} -free, then for every edge e and every vertex $x \notin e$ at most two of the four sets $T \cup \{x\}$ ($T \subset e$, $|T| = 3$) are edges, so $\sum_{T \subset e} \deg_G(T) \leq 2n - 4$ and $\text{co}_2(G) \leq (2n - 4)e(G)$ (de Caen’s double

count, as in the proof of [3, Theorem 8.3 of arXiv v3]); with $\pi(K_5^{4-}) \leq 1/2$ [28, Cor. 3.9] this gives $\sigma(K_5^{4-}) \leq 1/4$, which also follows from a one-block certificate on five vertices. The gaps to the lower bounds shrink from $9.5 \cdot 10^{-5}$ to $1.9 \cdot 10^{-6}$, from $6.9 \cdot 10^{-3}$ to $3.1 \cdot 10^{-4}$, and from 0.05 to 0.0040. For the other rows of Table 5 they shrink from $1.16 \cdot 10^{-3}$ to $3.64 \cdot 10^{-4}$ (C_5), from $1.88 \cdot 10^{-2}$ to $7.52 \cdot 10^{-3}$ ($K_6^{(3)}$), from $3.08 \cdot 10^{-2}$ to $2.65 \cdot 10^{-2}$ (K_5^-) and from $6.17 \cdot 10^{-2}$ to $4.80 \cdot 10^{-2}$ (K_5^{3-}); for K_6^{4-} , $K_6^{(4)}$ and $K_7^{(4)}$ we found no bounds in the literature beyond $\pi(F)^2 \leq \sigma(F) \leq \pi(F)$ (their gaps are now 0.111, $2.45 \cdot 10^{-2}$ and $4.27 \cdot 10^{-2}$). None of these values is claimed to be a seven-vertex optimum. The cut generation converged only for $K_6^{(4)}$ and $K_7^{(4)}$; for K_5^- and $K_6^{(3)}$ the certificates from cut generation stopped at its time limit, and for the other rows from the frozen-cut phase that followed such a stop. So, except for J_4 (Remark 26), the remaining gaps neither show nor rule out that seven vertices reach the lower bounds. For K_4^{3-} and for $\{K_4^{3-}, C_5\}$ the same method gives 0.0930631 and 0.0769412, within $3 \cdot 10^{-7}$ of, but not below, the bounds 0.09306286 and 0.07694083 of [3].

Remark 26. Seven vertices cannot reach $16/57$. An exact dual point on seven vertices, as in Section 6, is a probability vector y on 757 J_4 -free 7-vertex 3-graphs, with weights in $2^{-110}\mathbb{Z}$, whose moment matrices are all positive semidefinite (checked exactly, for all labelled types and all flag sizes with $2m - s \leq 7$). As in Lemma 28, every plain seven-vertex certificate for J_4 then has value $b \leq V = \sum_H y_H(1 - \gamma(H))$, with $1 - V = 0.2807032753\dots$, so it proves at best $\sigma(J_4) \leq 1 - V$: the plain seven-vertex method cannot prove any bound below 0.280703275315, and in particular misses $16/57$ by more than $1.5 \cdot 10^{-6}$. With Theorem 25, the plain seven-vertex optimum lies in $(0.280703275315, 0.280703690307]$. The obstruction is structural: for the law ϕ of the 7-vertex subgraph of the construction (93 classes, $\mathbb{E}_\phi(1 - \gamma) = 41/57$), a certificate of value $41/57$ would need $\mathbb{E}_\phi c_Q = 0$ for every block, and an exact rational witness on the support of ϕ shows that such certificates prove at best $\sigma(J_4) \leq 16/57 + \delta$, $\delta = 316891/14980607589 \approx 2.1 \cdot 10^{-5}$. As for Theorem 31, none of this covers stationarity constraints or more than seven vertices. For $K_5^{<}$ the run stopped before convergence, so it neither shows nor rules out that seven vertices reach $1/3$.

Remark 27. Two further lower bounds for 4-graphs: $\sigma(K_6^{4-}) > 0.5012201$, above the lower bound $\sigma(K_6^{4-}) \geq \sigma(K_5^{(4)}) = 31/64$, by a three-part pattern construction (parts of relative sizes 0.36827, 0.355115, 0.276615, whose edges are the 4-sets of patterns $(3, 0, 1)$, $(2, 2, 0)$, $(2, 1, 1)$, $(1, 2, 1)$, $(1, 1, 2)$, $(1, 0, 3)$, $(0, 3, 1)$; every six vertices span at most 13 edges); and $\sigma(K_6^{(4)}) > 0.7192198$ for the complement of Sidorenko's construction [42, Ex. 3.4] with equal Giraud halves and part weights optimised for σ .

6 What six and seven vertices can give

A *dual point on N vertices* is a probability vector $y = (y_H)$ on the admissible 4-graphs on N vertices (up to isomorphism) such that for every type σ of every pair (s, m) with $2m - s \leq N$ the moment matrix $Y_\sigma = \sum_H y_H M_\sigma(H)$, indexed by all admissible σ -flags on m vertices, is positive semidefinite.

Lemma 28. *If y is a dual point on N vertices, then every plain certificate on N vertices has value $b \leq \sum_H y_H d(H)$.*

Proof. $b \leq \sum_H y_H (d(H) - \sum c_Q(H)) = \sum_H y_H d(H) - \sum \langle Q', Y_\sigma \rangle$, where Q' is the principal submatrix of Q on the admissible flags of its list, extended by zero to all admissible flags (inadmissible flags never occur in an admissible graph). Each $\langle Q', Y_\sigma \rangle$ is nonnegative because Y_σ is symmetric (the two flags of a configuration play symmetric roles), so only the symmetric part of Q' matters, and both are positive semidefinite. $\square$

Proof of Theorem 9. At $N = 6$ the pairs are $(0, 1)$, $(0, 2)$, $(0, 3)$, $(1, 2)$, $(1, 3)$, $(2, 3)$, $(2, 4)$, $(3, 4)$ and $(4, 5)$, with two types for $(4, 5)$; pairs with $m = s$ have one flag per type, and their 1×1 moment matrices are nonnegative. The dual points of Table 6 have all ten moment matrices positive semidefinite and give $\sum_H y_H d(H) = 1/4, 1/10, 0$ and $1/2$. Lemma 28 completes the proof. $\square$

The positive semidefiniteness of the moment matrices was verified in exact arithmetic by two implementations, from the definitions (own flag enumeration, own densities, exact $LDL^\top$ test). For $t(5, 4)$ and $t(6, 4)$ all ten matrices have rank one; for $t(7, 4)$ nine have rank one and one is zero.

problem	support (pair graphs on $\{0, \dots, 5\}$)	weights	$\sum y_H d(H)$
$t(5, 4)$	perfect matching; star $K_{1,5}$; K_6	45/64, 9/32, 1/64	1/4
$t(6, 4)$	one pair; perfect matching; star $K_{1,5}$; K_6	81/100, 27/200, 27/500, 1/1000	1/10
$t(7, 4)$	the empty graph	1	0
$t_2(5, 4)$	$K_{1,1,4}$; C_6 ; two disjoint triangles; K_6	5/12, 5/12, 5/36, 1/36	1/2

Table 6: Dual points on six vertices. A 4-graph on six vertices is described by the graph of the complementary pairs of its edges; for instance “star $K_{1,5}$ ” is $K_5^{(4)}$ plus an isolated vertex.

Remark 29. On six vertices a 4-graph is a graph in disguise: the map $T \mapsto \{0, \dots, 5\} \setminus T$ sends 4-sets to pairs, and the five-vertex subsets correspond to the vertices. A 4-graph on six vertices is 5-admissible iff its pair graph has no isolated vertex, and 6-admissible iff it has an edge. This is why six vertices see so little of the problem. On seven vertices the same map sends 4-graphs to 3-graphs.

Remark 30. Six-vertex certificates of value exactly $1/4$, $1/10$ and $1/2$ exist; ours were checked exactly by the first implementation on all 2^{15} labelled 4-graphs on six vertices, and again by an independent checker of the data package written from the definitions (minimum over all admissible classes, attained on 3, 4 and 4 classes, the supports of the dual points). The lower bounds themselves are de Caen’s [10] and the general bound $\pi(H_k^r) \leq (k-2)/r$. So the plain six-vertex optimum is exactly de Caen’s bound for $K_5^{(4)}$ and $K_6^{(4)}$ and the $(k-2)/r$ bound for K_5^{4-} , and every improvement in Theorem 4(a)–(d) comes from seven-vertex flags. The statement is about the plain method only: Theorem 9 says nothing about six-vertex computations with additional constraints.

The barrier is not universal. For K_6^{4-} the plain six-vertex semidefinite program already gives about 0.19476 for $t_2(6, 4)$ (floating point, first implementation only), above the averaging bound 0.1476; seven vertices give the certified 0.21945 of Theorem 4(e).

The same weak-duality argument shows that, for $K_5^{(4)}$, the seven-vertex method is essentially exhausted.

Theorem 31. *No plain flag-algebra certificate on seven vertices proves*

$$t(5, 4) > V = \frac{1336237682928914994292138923}{7 \cdot 2^{89}} = 0.308401201194\dots,$$

that is, none proves $\pi(K_5^{(4)}) < 0.691598798805$. With Theorem 4(a), the best plain seven-vertex bound lies in $[0.308400990997, 0.308401201195]$, an interval of width less than $2.2 \cdot 10^{-7}$. The same holds for the problem restricted to odd 4-graphs.

Proof. At $N = 7$ the pairs (s, m) with $m > s$ and $2m - s \leq 7$ are $(0, 1)$, $(0, 2)$, $(0, 3)$, $(1, 2)$, $(1, 3)$, $(2, 3)$, $(1, 4)$, $(2, 4)$, $(3, 4)$, $(3, 5)$, $(4, 5)$ and $(5, 6)$; the pairs with $m = s$ are also allowed.

When $m \leq 3$ or $m = s$ there is one flag per type, so the moment matrix is a nonnegative 1×1 matrix. For the other six pairs we use the dual point y with weights $y_H = \nu_H/2^{110}$ (ν_H positive integers summing to 2^{110}) on 131 admissible 7-vertex 4-graphs, all of them odd. Every moment matrix Y_σ , for every labelled type σ of the six pairs, is positive semidefinite, and $\sum_H y_H d(H) = V$. Lemma 28 (which holds verbatim at $N = 7$) gives the first claim. Because every support graph is odd, the same y is a dual point for the odd-restricted problem. $\square$

The positive semidefiniteness was verified exactly by two independent implementations from the definitions. Rows and columns with zero diagonal vanish identically and are removed; the remaining parts of the (5, 6) matrices have dimensions 64, 58 and 64 (types with 1, 3 and 5 edges; types with 0, 2 or 4 edges do not occur in the support), and each is positive definite on its occurring flags, with a smallest normalised eigenvalue of about $1.4 \cdot 10^{-11}$. Moment matrices of blocks with smaller flags are obtained from these by summing flags, so the argument also covers certificates that mix flag sizes. The dual point was found by facial reduction of the restricted program's support followed by a primal-dual interior-point solve; that construction plays no role in the proof. The theorem does not cover stationarity constraints or more than seven vertices.

The same construction gives exact dual points for the other certificates of Theorem 4 (Table 7), again verified exactly by an independent implementation. For $t(7, 4)$ our bound is within $2.4 \cdot 10^{-6}$ of the best the plain seven-vertex method can give, for $t(6, 4)$ within $1.1 \cdot 10^{-5}$, and for $t_2(5, 4)$ and $t_2(6, 4)$ within $1.3 \cdot 10^{-3}$; in the last three cases the cut generation was stopped before it converged, and longer runs may close part of the gap. Without the (5, 6) block, that is, for certificates whose types of size five enter only through one-flag terms, no plain seven-vertex certificate proves $t(5, 4) > 0.281644555065$; so the (5, 6) block is worth at least 0.0267 in $t(5, 4)$.

problem	certified b (Theorem 4)	exact dual point V (rounded up)	$V - b$
$t(5, 4)$	0.308400990997	0.308401201195	$2.1 \cdot 10^{-7}$
$t(6, 4)$	0.139827122482	0.139837689411	$1.1 \cdot 10^{-5}$
$t(7, 4)$	0.064762076605	0.064764464150	$2.4 \cdot 10^{-6}$
$t_2(5, 4)$	0.552992751369	0.553946861054	$9.5 \cdot 10^{-4}$
$t_2(6, 4)$	0.219452823117	0.220748423194	$1.3 \cdot 10^{-3}$
$t(5, 4)$, no (5, 6) block	—	0.281644555065	—

Table 7: What the plain seven-vertex method can give: no plain seven-vertex certificate proves more than V . The supports of the dual points are odd only for $t(5, 4)$ (Theorem 31).

For comparison, at $r = 3$ the plain method on $r + 2 = 5$ vertices already beats the bound of Chung and Lu for $K_4^{(3)}$: it gives $t(4, 3) > 0.426923$, that is $\pi(K_4^{(3)}) < 0.573077$, against $\pi(K_4^{(3)}) \leq (3 + \sqrt{17})/12 < 0.593593$ [8]. The first implementation found this value; an independent script written from the definitions (in the data package [30]) gives the exact certificate $t(4, 3) \geq 60084175574225/2^{47} > 0.4269237$ over all 768 labelled admissible 3-graphs on five vertices. And for $r = 5$ the seven-vertex computation of Theorem 5 beats it for $K_6^{(5)}$; among $r = 3, 4, 5$, only for $r = 4$ does the $(r + 2)$ -vertex method stop at de Caen's bound.

7 Remarks on the seven-vertex optimum

Except where marked exact, the observations in this section are floating-point computations; none is used in a proof.

Which blocks matter. For $t(5, 4)$, a run with all blocks except $(5, 6)$ converges to 0.28164; a run with the $(5, 6)$ block alone reaches 0.30787 before its time limit; a run with only $(3, 5)$ and $(5, 6)$ reaches 0.30840117; the full run gives 0.30840112. In the certificates of Theorem 4 the blocks $(2, 4)$ and $(3, 4)$ are zero, $(1, 4)$ is zero except for $t_2(6, 4)$ (rank one), and $(4, 5)$ has rank one or zero (Appendix A); the sharp certificate of Theorem 6 also uses $(1, 4)$ and $(4, 5)$.

Distance to the seven-vertex optimum. The exact interval is given in Theorem 31. In floating point the gap is smaller: the frozen-cut value is an optimum over an inner approximation of the positive semidefinite cone, and the $(3, 5) + (5, 6)$ run above exceeds it by $5.6 \cdot 10^{-8}$, so the optimum is at least 0.3084011746 in floating point and lies in $[0.3084011746, 0.3084012012]$. Adding the stationarity constraints of the differential method (degree regularity of extremal graphs) raises the floating-point value by about $2.7 \cdot 10^{-6}$. By our own analysis, not independently reviewed, at seven vertices these constraints reduce to a single scalar condition, because the rooted edge flag already has four vertices. We did not certify this gain.

The optimum is odd. Recall that a 4-graph is odd if every five vertices span an odd number of edges (Section 4). Giraud’s construction is odd (Lemma 3). The dual point of Theorem 31 is supported on odd graphs (exact), and the optimal pseudo-density of the final restricted program is odd up to $5.8 \cdot 10^{-6}$ (float): in both, a random five-set spans 1, 3 or 5 edges with probabilities about 0.7714, 0.1862 and 0.0424, against $25/32$, $5/32$ and $1/16$ for Giraud’s construction. The program finds the parity structure of Giraud’s construction although nothing in it mentions parity. The certificate itself does not force parity (exact): its minimum b is attained at a graph with fourteen even five-sets, while every odd 7-vertex 4-graph (there are 423 classes) has $d - \sum c_Q \geq b + 376193/28862180229120$. Restricted to odd 4-graphs, the certificate of Theorem 4(a) itself is worth $b + 376193/28862180229120 > 0.3084010040$, by the margin just stated, and no plain certificate is worth more than V (the dual point of Theorem 31 is supported on odd graphs); at seven vertices parity is not the missing information.

Remark 32. A 4-graph A on V is odd iff $A = \{T : T \text{ contains an even number of edges of } J\}$ for some 3-graph J on V . Indeed, a 4-set is a 3-simplex, and A is odd iff the indicator of the complement of A is a 3-cocycle of the full simplex on V over $\mathbb{F}_2$; every such cocycle is the coboundary of a 2-cochain, the indicator of a 3-graph J . So the question whether every odd 4-graph has density at least $5/16 - o(1)$, a weak form of Sidorenko’s conjecture, is a question about 3-graphs.

Giraud’s construction against the certificate (exact). Take Giraud’s construction with equal parts and independent uniform matrix entries. Its random 7-vertex subgraph has an explicit law: each vertex lies in a uniformly random part, and the entries of the matrix on the sampled rows and columns are independent uniform bits. The law is supported on 2404 labelled graphs in 10 isomorphism classes, all odd. For the certificate of Theorem 4(a) we computed exactly $\mathbb{E} d = 5/16$, $\mathbb{E}(d - \sum c_Q) = 173614396936021/2^{49} > 0.3084011214$ and $\mathbb{E} \sum c_Q = 2307463508139/2^{49} > 0.0040988785$, and every support graph satisfies $b + 7.09 \cdot 10^{-8} \leq d - \sum c_Q \leq b + 4.53 \cdot 10^{-7}$. A Giraud system of density $5/16 + o(1)$ has parts of size $n/2 + o(n)$ and a fraction $1/2 + o(1)$ of even minors (Lemma 3). Every Giraud system on seven vertices lies in the support of this law ((G7) in Section 4), so every 7-vertex subgraph of every Giraud system is tight for the certificate up to $4.6 \cdot 10^{-7}$, and the remaining gap $5/16 - b$ exceeds the sum-of-squares term of the law by about $1.3 \cdot 10^{-7}$. Closing the gap needs larger flags or additional structure, such as stability. For the codegree-squared objective the same law is tight exactly (Section 3.8).

Eight vertices are out of reach. About 23% of the 2^{70} labelled 4-graphs on eight vertices are 5-admissible (Monte Carlo), which gives about $6.7 \cdot 10^{15}$ classes. The new block (6, 7) would have 122 types with up to 2^{20} flags each. Any eight-vertex certificate, even one without the new block, is a minimum over all admissible 8-vertex graphs, so it needs an exact scan of about 10^{16} classes or of the $1.3 \cdot 10^{17}$ one-vertex extensions of the seven-vertex classes; we estimate a single scan at least three orders of magnitude beyond a modest cloud budget.

8 Consequences

8.1 Lottery numbers

An (n, k, r, p) -lottery system is a family of k -subsets (tickets) of an n -set such that every p -subset (draw) meets some ticket in at least r points; $L(n, k, r, p)$ is its least size. We follow Sidorenko's order of arguments [43]; the lotto tables [25] write $L(n, k, p, r)$, so their $L(n, 6, 6, 4)$ is $L(n, 6, 4, 6)$ here. Sidorenko proved that $\ell(k, r, p) = \lim_n L(n, k, r, p) / \binom{n}{r}$ exists [43].

Lemma 33 (Shadow bound [4, Lemma 4.1.1], [43, eq. (2)]). *The r -subsets of the tickets of an (n, k, r, p) -lottery system form a Turán (n, p, r) -system. Hence $L(n, k, r, p) \geq T(n, p, r) / \binom{k}{r}$ and $\ell(k, r, p) \geq t(p, r) / \binom{k}{r}$.*

Proof. A draw that meets a ticket in at least r points contains an r -subset of that ticket, and a ticket has $\binom{k}{r}$ such subsets. $\square$

Corollary 34. *For every $k \geq 4$,*

$$\begin{aligned} \binom{k}{4} \ell(k, 4, 5) &> 0.308400, & \binom{k}{4} \ell(k, 4, 6) &> 0.139827, \\ \binom{k}{4} \ell(k, 4, 7) &> 0.064762. \end{aligned}$$

In particular $L(n, 6, 4, 6) \geq (1 - o(1)) \binom{n}{4} / 107.28$.

Table 8 gives the cases $5 \leq k, p \leq 7$. These are asymptotic statements; they do not improve entries of the finite tables [25] directly. The previous lower bounds are the same shadow bound with the previous values of $t(p, 4)$. The new lower bounds remove between 13.2% (for $p = 7$) and 32.3% (for $(5, 4, 5)$) of the gaps between the previous lower bounds and the upper bounds of the table, measured on the normalised constants $\binom{k}{4} \ell(k, 4, p)$; the upper bounds for $(6, 4, 5)$, $(7, 4, 5)$ and $(7, 4, 6)$ are those of Proposition 45 (Section 8.4).

8.2 Covering insertion codes

A set $C \subseteq [n]^r$ is a *covering $(k - r)$ -insertion code* if every word in $[n]^k$ contains a member of C as a subsequence; let $S(n, k, r)$ be its least size and $s(k, r) = \lim_n S(n, k, r) / n^r$. Pikhurko, Verbitsky and Zhukovskii proved $s(k, r) = t(k, r)$ [36, Theorem 18 of arXiv v3].

Corollary 35. $s(5, 4) > 0.308400$, $s(6, 4) > 0.139827$, $s(7, 4) > 0.064762$, $s(6, 5) > 0.232599$ and $s(7, 5) > 0.072082$.

8.3 Giraud systems are expensive to cover

Lemma 33 gives $\binom{k}{4} \ell(k, 4, 5) \geq t(5, 4)$, and we do not know whether equality can hold for some $k \geq 5$. If Sidorenko's conjecture $t(5, 4) = 5/16$ holds, equality would require lottery systems whose shadows are near-extremal Turán systems, and the best known constructions of Turán $(n, 5, 4)$ -systems are Giraud's [41, Section 8], [42, Section 2]. We show that these are far from being shadows of cheap lottery systems.

$(k, 4, p)$	this paper	previous	upper bound	source of the upper bound
$(5, 4, 5)$	16.22	18.92	12.48	[32, Cor. 5.6]
$(6, 4, 5)$	48.64	56.74	34.28	Prop. 45(a)
$(7, 4, 5)$	113.49	132.39	78.23	Prop. 45(b)
$(5, 4, 6)$	35.76	47.43	20.44	[32, Thm 5.9]
$(6, 4, 6)$	107.28	142.28	58.24	[32, Section 6], from [31]
$(7, 4, 6)$	250.31	331.98	124.05	Prop. 45(c)
$(5, 4, 7)$	77.21	89.91	40	[32, eq. (12)]
$(6, 4, 7)$	231.62	269.72	120	[32, eq. (12)]
$(7, 4, 7)$	540.44	629.33	280	[32, eq. (12)]

Table 8: Asymptotic bounds on $L(n, k, 4, p)$, written as $\binom{n}{4}/x$ up to a factor $1 \pm o(1)$; the table gives x , rounded up for the lower bounds and down for the upper bounds. Previous lower bounds use $t(5, 4) > 0.264386$, $t(6, 4) > 0.105430$ and $t(7, 4) > 0.055614$, by repeated averaging from Sidorenko's table values [42] as in Table 1. Montecalvo's $\delta(k, t, p)$ is $\binom{p}{t}$ times our normalised constant $\binom{k}{t}\ell(k, t, p)$; the value for $(6, 4, 6)$, $\delta < 3.863$, is quoted in [32] from [31]. The previous upper bounds for the rows of Proposition 45 were 33.21 for $(6, 4, 5)$ ($\delta \leq 2.258$, quoted in [32, Section 5.2] from [31]), and 74.60 and 122.84 for $(7, 4, 5)$ and $(7, 4, 6)$ [32, Thm 5.12], a theorem that assumes that the limits exist, which Sidorenko proved [43].

Recall the Giraud systems of Definition 2 and their density bound in Lemma 3. For a 4-graph Γ on $[n]$ and $k \geq 5$ let $\tau_k(\Gamma)$ be the fractional cover number of $E(\Gamma)$ by k -subsets of $[n]$, the least $\sum_K w_K$ over weights $w \geq 0$ on k -sets with $\sum_{K \supseteq T} w_K \geq 1$ for every edge T . Since a k -set contains $\binom{k}{4}$ quadruples, $\binom{k}{4}\tau_k(\Gamma) \geq e(\Gamma)$.

Theorem 36. *For every $k \geq 5$ and every Giraud system $\Gamma = \Gamma(A, B, M)$ on n vertices, with $s = |A||B|/n^2$,*

$$\binom{k}{4} \frac{\tau_k(\Gamma)}{\binom{n}{4}} \geq 1 - 4s + 7s^2 - O(1/n) \geq \frac{7}{16} - O(1/n),$$

uniformly in M and the part sizes. The density of Γ can be as small as $5/16 + o(1)$.

Lemma 37. *Let $k \geq 5$, $i + j = k$ and $i, j \geq 0$. Then $\binom{i}{4} + \binom{j}{4} + \frac{5}{3}\binom{i}{2}\binom{j}{2} \leq \binom{k}{4}$.*

Proof. By Vandermonde, $\binom{k}{4} = \binom{i}{4} + \binom{i}{3}j + \binom{i}{2}\binom{j}{2} + i\binom{j}{3} + \binom{j}{4}$, so the claim is $\frac{2}{3}\binom{i}{2}\binom{j}{2} \leq \binom{i}{3}j + i\binom{j}{3}$. It is trivial if $i \leq 1$ or $j \leq 1$. Otherwise divide by $ij/6$ and put $a = i - 1$, $b = j - 1 \geq 1$; the claim becomes $ab \leq a(a - 1) + b(b - 1)$, that is $(a - b)^2 + (a - 1)(b - 1) - 1 \geq 0$. If $a \neq b$ the first term is at least 1. If $a = b$, then $2a = k - 2 \geq 3$, so $a \geq 2$ and $(a - 1)^2 \geq 1$. $\square$

Proof of Theorem 36. If a part has fewer than two points, then $s = O(1/n)$, Γ contains all but $O(n^3)$ quadruples, and the claim follows from $\binom{k}{4}\tau_k(\Gamma) \geq e(\Gamma)$. Otherwise, by LP duality, $\tau_k(\Gamma) \geq \sum_T y_T$ for every $y \geq 0$ on $E(\Gamma)$ with $\sum_{T \subseteq K} y_T \leq 1$ for every k -set K . Give weight $y_I = 1/\binom{k}{4}$ to the internal edges and $y_X = \frac{5}{3}/\binom{k}{4}$ to the cross edges. A k -set with i points in A and j in B contains exactly $\binom{i}{4} + \binom{j}{4}$ internal edges and at most $\binom{i}{2}\binom{j}{2}$ cross edges, whatever M is; so Lemma 37 says that y is feasible. Hence

$$\binom{k}{4}\tau_k(\Gamma) \geq \binom{|A|}{4} + \binom{|B|}{4} + \frac{5}{3}f\binom{|A|}{2}\binom{|B|}{2},$$

and Lemma 3 gives $1 - 4s + 2s^2 + 10s^2f - O(1/n) \geq 1 - 4s + 7s^2 - O(1/n)$ after division by $\binom{n}{4}$; the error term from f is $O(1/n)$ because $\max(|A|, |B|) \geq n/2$. Finally $1 - 4s + 7s^2$ is decreasing on $[0, 1/4]$, with value $7/16$ at $s = 1/4$. $\square$

Remark 38. For $k = 5, 6$ no other choice of the two weights gives more. Optimising the two weights for each k at the extremal members ($s = 1/4$, $f = 1/2$) gives the exact values $7/16$, $7/16$, $23/48$, $23/48$, $1/2$, $1/2$, $41/80$, $41/80$ for $k = 5, \dots, 12$; over all part sizes the minimum is the same for $k \leq 10$ and $21/41$ (at $s = 10/41$) for $k = 11, 12$. For $k = 5$ the proof is one line: a five-set containing a cross edge has at most three edges, all cross, so $\tau_5 \geq |E_{\text{int}}|/5 + |E_{\text{cross}}|/3$. For $k = 6$ the bound $7/16$ of Theorem 36 is sharp. The weights in the proof of Proposition 45(a) below are a fractional cover by 6-sets of the edges of Γ_d , which is Giraud's construction in the sense of this section (equal parts, $f \rightarrow 1/2$), so $\binom{6}{4}\tau_6(\Gamma_d)/\binom{n}{4} \leq 15W_N/\binom{n}{4} \rightarrow 7/16$; by Lemma 43 there is even an integral cover of the same asymptotic size, which is an $(n, 6, 4, 5)$ -lottery system.

For 4-graphs Γ, Γ' on $[n]$ write $\Gamma \setminus \Gamma'$ for the set difference of the edge sets. The shadow ∂G of a lottery system G is the 4-graph of the 4-subsets of its tickets.

Corollary 39. *Let G be an $(n, k, 4, 5)$ -lottery system and Γ a Giraud system on the same n points with $|\Gamma \setminus \partial G| \leq \eta n^4$. Then $\binom{k}{4}|G|/\binom{n}{4} \geq 7/16 - 24\binom{k}{4}\eta - o(1)$.*

Proof. The tickets of G , together with one k -set through each edge of $\Gamma \setminus \partial G$, cover every edge of Γ . So $\tau_k(\Gamma) \leq |G| + \eta n^4$, and $n^4/\binom{n}{4} \rightarrow 24$. $\square$

Corollary 40. *If G_n are $(n, 5, 4, 5)$ -lottery systems of size $L(n, 5, 4, 5)$, then every Giraud system Γ on the points of G_n has $|\Gamma \setminus \partial G_n| \geq (1/3240 - o(1))n^4 = (1/135 - o(1))\binom{n}{4}$.*

Proof. Montecalvo [32, Cor. 5.6] gives $5L(n, 5, 4, 5)/\binom{n}{4} \leq 865/2160 + o(1)$. With $k = 5$, Corollary 39 forces $120\eta \geq 7/16 - 865/2160 - o(1) = 1/27 - o(1)$. $\square$

So optimal $(n, 5, 4, 5)$ -lottery systems do not have shadows close to Giraud systems. For $k = 6$ Theorem 36 gives no such statement: the lottery systems in the proof of Proposition 45(a) (Section 8.4) cover every edge of Γ_d and attain its bound $7/16$. Instead, Corollary 39 says that an $(n, 6, 4, 5)$ -lottery system G with $\binom{6}{4}|G|/\binom{n}{4} \leq 7/16 - \varepsilon$ has $|\Gamma \setminus \partial G| \geq (\varepsilon/360 - o(1))n^4$ for every Giraud system Γ , so any improvement of Proposition 45(a) must have a shadow far from every Giraud system. With a stability hypothesis, Corollary 39 gives a strict inequality in the shadow bound for every k (Corollary 42).

Hypothesis 41 (S_G). For every $\varepsilon > 0$ there are $\delta > 0$ and n_0 such that every Turán $(n, 5, 4)$ -system with $n \geq n_0$ and density at most $5/16 + \delta$ satisfies $|\Gamma \setminus G| \leq \varepsilon n^4$ for some Giraud system Γ on the same vertex set.

Hypothesis S_G implies Sidorenko's conjecture $t(5, 4) = 5/16$, because a Giraud system has density at least $5/16 - o(1)$. It is a stability form of that conjecture. Markström conjectured that for $n \geq 12$ the only extremal $K_5^{(4)}$ -free 4-graphs are those of Giraud's construction and its generalisation [27, Conjecture 4.2], and he wrote that he expects "a stability version of Conjecture 4.2 ... showing that the near-extremal hypergraphs are also close to Giraud's construction" [27, Section 4.1]. Hypothesis S_G is a version of this, in complement form, that allows any Giraud system.

Corollary 42. *If Hypothesis S_G holds, then $\binom{k}{4}\ell(k, 4, 5) > 5/16 \geq t(5, 4)$ for every $k \geq 5$: the shadow bound is not tight for $(k, 4, 5)$.*

Proof. Put $\varepsilon = 1/(384\binom{k}{4})$ and let δ be given by S_G . Let G be an optimal lottery system. If $d(\partial G) > 5/16 + \delta$, then $\binom{k}{4}|G|/\binom{n}{4} \geq d(\partial G) > 5/16 + \delta$. Otherwise ∂G is a Turán $(n, 5, 4)$ -system (Lemma 33) of density at most $5/16 + \delta$, and Corollary 39 gives $\binom{k}{4}|G|/\binom{n}{4} \geq 7/16 - 24\binom{k}{4}\varepsilon - o(1) = 3/8 - o(1)$. In both cases the limit exceeds $5/16$, and $t(5, 4) \leq 5/16$ by Giraud's construction. $\square$

8.4 Upper bounds for lottery numbers

We improve three upper bounds of Table 8, Montecalvo's normalised constants 0.4516, 38/81 and 7385/25920 [32], with fractional covers made integral by Kahn's fractional form [22] of the Frankl–Rödl–Pippenger theorem [16, 37]. For a hypergraph X on a finite set V , a map $\varphi: X \rightarrow \mathbb{R}_{\geq 0}$ is a *fractional cover* if $\sum_{A \ni v} \varphi(A) \geq 1$ for every $v \in V$; let $\varphi(X) = \sum_{A \in X} \varphi(A)$ and $\alpha(\varphi) = \max_{u \neq v} \sum_{A \ni u, v} \varphi(A)$, and let $\rho(X)$ be the least number of edges of X whose union is V . Kahn [22, Theorem 1.2(b)] proved: for fixed h , if every edge of X has at most h elements and φ is a fractional cover, then $\rho(X) \leq (1 + o(1)) \varphi(X)$ as $\alpha(\varphi) \rightarrow 0$, uniformly in X . (He derives it, for $\varphi(X) \rightarrow \infty$, from his Theorem 1.4 on fractional matchings, applied to a *fractional tiling*, where $\sum_{A \ni v} \varphi(A) = 1$ for every v . Moving, vertex by vertex, the excess weight at v from edges $A \ni v$ to $A \setminus \{v\}$ (deleting it if $A = \{v\}$) turns a fractional cover into a tiling of the down-closure of X with no larger $\varphi(X)$ or $\alpha(\varphi)$, and a cover of the down-closure gives one of X of the same size. In part (a) of Proposition 45, φ is already a tiling.)

Lemma 43. *Fix $k \geq r$. For n in an infinite set let G_n be an r -graph on $[n]$ with $e(G_n) \rightarrow \infty$, and $w_n \geq 0$ weights on the k -subsets of $[n]$ with $\sum_{K \supseteq R} w_n(K) \geq 1$ for every edge R of G_n and $\alpha_n = \max_{R \neq R'} \sum_{K \supseteq R \cup R'} w_n(K) \rightarrow 0$, the maximum over distinct edges R, R' of G_n . Then there is a family of at most $(1 + o(1)) \sum_K w_n(K)$ k -sets such that every edge of G_n lies in one of them. If G_n is a Turán (n, p, r) -system, this family is an (n, k, r, p) -lottery system.*

Proof. Let X_n be the hypergraph on $E(G_n)$ whose edges are the nonempty sets $E_K = \{R \in E(G_n) : R \subseteq K\}$ with $w_n(K) > 0$, and put $\varphi(E) = \sum_{K: E_K = E} w_n(K)$. Its edges have at most $\binom{k}{r}$ elements, φ is a fractional cover, $\alpha(\varphi) = \alpha_n$ and $e(G_n)/\binom{k}{r} \leq \varphi(X_n) \leq \sum_K w_n(K)$. Kahn's theorem gives a cover of X_n by $(1 + o(1)) \varphi(X_n)$ sets E_K ; take their k -sets K . A p -set contains an edge of a Turán (n, p, r) -system, and hence meets the k -set containing that edge in at least r points. $\square$

For type patterns fix $m \geq 1$ and weights $x \in (0, 1]^m$ with $\sum_i x_i = 1$. A *type* is a vector $e \in \mathbb{Z}_{\geq 0}^m$ of size $|e| = \sum_i e_i$; for a partition $[n] = V_1 \cup \dots \cup V_m$, the type of a set S is $(|S \cap V_i|)_{i \leq m}$. For types e of size r and B of size k let $a(e, B) = \prod_i \binom{B_i}{e_i}$, the number of r -subsets of type e of a k -set of type B , and $\mu_e(x) = r! \prod_i x_i^{e_i} / e_i!$. We write $P \geq e$ coordinatewise.

Lemma 44. *Let $k \geq r$, $p \geq r$ and $q \geq r$, let $I \subseteq [m]$, and let $\mathcal{T}$ be a nonempty set of types of size r such that*

- (i) *every type P of size p with $P_i < q$ for all $i \in I$ satisfies $P \geq e$ for some $e \in \mathcal{T}$, and*
- (ii) *some $y = (y_B) \geq 0$, indexed by the types B of size k , satisfies $\sum_B a(e, B) y_B \geq \mu_e(x)$ for every $e \in \mathcal{T}$.*

Then $\ell(k, r, p) \leq \sum_B y_B + \ell(k, r, q) \sum_{i \in I} x_i^r$.

Proof. Split $[n]$ into parts V_i with $|V_i| = x_i n + O(1)$ and let G_n consist of the r -sets with type in $\mathcal{T}$. There are $N_e = \prod_i \binom{|V_i|}{e_i} = (1 + O(1/n)) \mu_e(x) \binom{n}{r}$ r -sets of type e and $N_B = \prod_i \binom{|V_i|}{B_i} = \Theta(n^k)$ k -sets of type B , and an r -set of type e lies in $a(e, B) N_B / N_e$ of the latter. Give each k -set of type B the weight $\lambda y_B \binom{n}{r} / N_B$, where $\lambda = \max_{e \in \mathcal{T}} N_e / (\mu_e(x) \binom{n}{r}) = 1 + O(1/n)$. By (ii) every $R \in G_n$ of type e gets total weight $\lambda \binom{n}{r}^{-1} \sum_B a(e, B) y_B \geq 1$; the total weight is $\lambda \binom{n}{r} \sum_B y_B$; and $\alpha_n = O(1/n)$, because two distinct r -sets span at least $r + 1$ points, which lie in $O(n^{k-r-1})$ k -sets, each of weight $O(n^{r-k})$. Lemma 43 gives $(1 + o(1)) \binom{n}{r} \sum_B y_B$ k -sets that contain all members of G_n . Add, for each $i \in I$, an optimal $(|V_i|, k, r, q)$ -lottery system on V_i ; it has $(\ell(k, r, q) x_i^r + o(1)) \binom{n}{r}$ tickets, since $\ell(k, r, q)$ is a limit [43]. Every p -set P meets some ticket in at least r points: if $|P \cap V_i| \geq q$ for some $i \in I$, a ticket of the system on V_i ; otherwise, by (i), P contains an r -set with type in $\mathcal{T}$, which lies in one of the k -sets above. Divide by $\binom{n}{r}$ and let $n \rightarrow \infty$. $\square$

Proposition 45. (a) $\binom{6}{4}\ell(6, 4, 5) \leq 7/16$; that is, $L(n, 6, 4, 5) \leq (1 + o(1))7\binom{n}{4}/240 \leq (1 + o(1))\binom{n}{4}/34.28$.

(b) $\binom{7}{4}\ell(7, 4, 5) \leq c_{745}$, where

$$c_{745} = \frac{3467292853891919529774577}{7750000000000000000000000} < 0.447393.$$

(c) $\binom{7}{4}\ell(7, 4, 6) \leq (107 + 3c_{745})/384 < 0.282142$.

Proof. (a) Let $d \geq 4$, $N = 2^d$ and $n = 2N$, let A and B be disjoint copies of $\mathbb{F}_2^d$, and let Γ_d be the Giraud system $\Gamma(A, B, M)$ of Definition 2 with $M_{ab} = \langle a, b \rangle$, the standard bilinear form over $\mathbb{F}_2$. The minor of $\{a, a', b, b'\}$ is $\langle a + a', b + b' \rangle$, so for fixed $a \neq a'$ a fraction $(N/2 - 1)/(N - 1)$ of the pairs $\{b, b'\}$ give cross edges and $f \rightarrow 1/2$; by Lemma 3, Γ_d is a Turán $(n, 5, 4)$ -system with equal parts. Call a 6-set *good* if it lies in A or in B , or if it is a set $\{a_1, a_2, a_3, b_1, b_2, b_3\}$ whose nine cross 4-subsets are edges. With $u = a_1 + a_2$, $v = a_1 + a_3$, $w = b_1 + b_2$ and $z = b_1 + b_3$, the differences inside the two triples are $u, v, u + v$ and $w, z, w + z$, so by bilinearity the set is good iff $\langle u, w \rangle = \langle u, z \rangle = \langle v, w \rangle = \langle v, z \rangle = 0$. Given a cross edge $\{a_1, a_2, b_1, b_2\}$, so that $u, w \neq 0$ and $\langle u, w \rangle = 0$, the good sets containing it correspond to the pairs (v, z) with $v \in w^\perp \setminus \{0, u\}$, which leaves $N/2 - 2$ choices because $u \in w^\perp$, and $z \in \{u, v\}^\perp \setminus \{0, w\}$, which leaves $N/4 - 2$ choices because u, v are linearly independent and $w \in \{u, v\}^\perp$. So every cross edge lies in exactly $c_N = (N/2 - 2)(N/4 - 2)$ good sets, and every internal edge in $\binom{N-4}{2}$ good sets inside its part. The weights $1/c_N$ on the good sets of type $(3, 3)$ and $1/\binom{N-4}{2}$ on the other good sets give every edge of Γ_d total weight exactly 1, and since a good set contains 9 cross or 15 internal edges, the total weight is

$$W_N = \frac{|E_{\text{cross}}|}{9} + \frac{2\binom{N}{4}}{15}, \quad |E_{\text{cross}}| = \binom{N}{2} \left(\frac{N}{2} - 1 \right) \frac{N}{2},$$

so that $15W_N/\binom{n}{4} \rightarrow 15\left(\frac{1}{72} + \frac{1}{180}\right)\frac{3}{2} = \frac{7}{16}$. Two distinct edges in a common good set span at least five of its points, which leaves at most N choices for the sixth, and no good set contains both an internal and a cross edge; so $\alpha_n \leq N \max(1/c_N, 1/\binom{N-4}{2}) = O(1/N)$. Lemma 43 gives $L(n, 6, 4, 5) \leq (1 + o(1))W_N$ along $n = 2^{d+1}$, and $\ell(6, 4, 5)$ is a limit [43].

(b) This is Lemma 44 with $m = 5$ and $I = \emptyset$ (so q plays no role), the part weights $x_1 = 0.317221$, $x_2 = x_4 = 0.126888$, $x_3 = 0.302114$ and $x_5 = 0.126889$, a set $\mathcal{T}$ of 38 types (the five types with four points in one part, ten of shape $(3, 1)$, ten of shape $(2, 2)$, twelve of shape $(2, 1, 1)$ and one of shape $(1, 1, 1, 1)$) and 27 positive rational values y_B with $\binom{7}{4} \sum_B y_B = c_{745}$, listed in the data package [30]. Conditions (i) (on the 126 types of size 5) and (ii) (38 rational inequalities, 27 of them tight) were checked in exact arithmetic by two independent programs, and (i) also by brute force on a blow-up with five points in each part.

(c) This is Lemma 44 with $m = 4$, $x_i = 1/4$, $q = 5$ and $I = \{1, 2\}$. Write $i \rightarrow j$ for an ordered pair of parts and use the six arcs $1 \rightarrow 2$, $2 \rightarrow 3$, $3 \rightarrow 4$, $4 \rightarrow 1$, $1 \rightarrow 3$, $2 \rightarrow 4$, one on each pair of parts. $\mathcal{T}$ consists of the types (4) inside parts 3 and 4, the six types $(2, 2)$, and for each arc $i \rightarrow j$ the type with three points in part i and one in part j . Put $y_B = 1/13440$ for the 7-sets inside part 3 and for those inside part 4, and $y_B = 1/768$ for the six block types with four points in part i and three in part j , one for each arc $i \rightarrow j$. For (i), let P have size 6 and $P_1, P_2 \leq 4$. If $P_3 \geq 4$, or $P_4 \geq 4$, or two parts have at least two points each, P is at least a type of $\mathcal{T}$. Otherwise some part i has $P_i \geq 3$ and every other part at most one point. If $P_i = 3$, the other three parts have a point each and i is the tail of an arc; if $P_i = 4$, then $i \in \{1, 2\}$, two of the other parts have a point, and i is the tail of two arcs $(1 \rightarrow 2, 3, \text{ resp. } 2 \rightarrow 3, 4)$, one of which ends in such a part. For (ii), at $x_i = 1/4$ we have $\mu_e = 1/256$, $3/128$ and $1/64$ for the types (4), $(2, 2)$ and $(3, 1)$. The 4-sets inside part 3 occur only in the 7-sets inside part 3, 35 in

each, and in the blocks of the arc $3 \rightarrow 4$, one in each: $35/13440 + 1/768 = 1/256$; the same holds for part 4 with the arc $4 \rightarrow 1$. A type $(2, 2)$ occurs only in the blocks of the arc on its pair, $\binom{4}{2}\binom{3}{2} = 18$ times in each, and the type $(3, 1)$ of an arc only in the blocks of that arc, $\binom{4}{3}\binom{3}{1} = 12$ times in each: $18/768 = 3/128$ and $12/768 = 1/64$. So all 14 rows are tight, and $\binom{4}{4} \sum_B y_B = 35(2/13440 + 6/768) = 107/384$. The second term of Lemma 44, multiplied by $\binom{7}{4}$, is $\binom{7}{4} \ell(7, 4, 5)(x_1^4 + x_2^4) \leq c_{745}/128$ by (b), and $107/384 + c_{745}/128 = (107 + 3c_{745})/384$. $\square$

The pattern in (c) has the shape of Montecalvo’s bound [32, Theorem 5.10(3)]. With his value $38/81$ for $(7, 4, 5)$ in place of c_{745} it gives $(107 + 3 \cdot \frac{38}{81})/384$, exactly $1/384$ below his $7385/25920$; this gain comes from the rows of type (4), to which the blocks with four points in parts 3 and 4 also contribute.

9 Data and reproducibility

The accompanying data package [30] contains all certificates, independent verifiers, a driver that recomputes every exact value, and the search code. Checking the certificates does not require the search code. Code is under the MIT licence; data are under CC BY 4.0.

For the five 4-graph bounds, each certificate contains the block and flag lists, integer matrices A , scale M and claimed fraction; together they occupy about 4.1 MB. The sharp certificate of Theorem 6 stores W , B , X and the common denominator $D = 57 \cdot 2^{152}$. The package also contains the ten threshold certificates, the ten further codegree-squared certificates with their type and flag lists, the three 5-graph certificates, the witness and dual point of Remark 26, the six- and seven-vertex dual points, and the two type-pattern certificates of Proposition 45(b) and (c).

To check a bound of Theorem 4, a verifier generates the 6-vertex representatives in seconds and scans the raw extensions in Table 4. This takes a few minutes per certificate with the first implementation’s compiled evaluator on two cores, or about ten minutes to three and a half hours with the independent C evaluator (Section 3.6). The class lists (137 MB) are not needed for the proof. The package’s `verifier/verify_all.sh -fast` and `-raw all` checks were run on clean cloud machines before release; all passed.

AI disclosure

AI coding agents, built on several models (including Claude and GPT), assisted the implementation, the computations, the proofs and the writing of this paper. The search code that found the certificates and the verifiers that re-checked them were written in separate agent sessions that did not share code, and every result was checked by a separate agent session before it entered the paper. The cutting-plane and column-generation scheme follows Jeong et al. [21], in an independent implementation that does not use their code. The author ran the fast checks of the data package (`verify_all.sh -fast`); all its checks, including the exhaustive scans, were also run on clean cloud machines.

A The certificates

Table 9 lists the parameters of the five 4-graph certificates. In every case the stationarity weights are zero. “Rank” is the number of rows of A per block, in the order of Table 3; the blocks $(2, 4)$ and $(3, 4)$ are zero throughout and are omitted, and $(1, 4)$ has rank one for $t_2(6, 4)$ and zero otherwise.

	$t(5, 4)$	$t(6, 4)$	$t(7, 4)$	$t_2(5, 4)$	$t_2(6, 4)$
M	2^{20}	2^{21}	2^{20}	2^{21}	2^{21}
rank (3, 5)	11	12	11	9	16
rank (4, 5)	1, 1	0, 0	0, 0	0, 0	0, 0
rank (5, 6)	103, 119, 91, 33, 35	64, 182, 194, 116, 54, 21	35, 94, 73, 58, 41, 27	189, 225, 201, 124	77, 214, 280, 133, 77, 51
float LP	0.30840111889	0.13982724282	0.06476227826	0.55299286856	0.21945291606
certified b	0.308400990997	0.139827122482	0.064762076605	0.552992751369	0.219452823117
minimiser	15 edges	co-Fano	13 edges	25 edges	12 edges

Table 9: Parameters of the certificates (certified values floored). “Co-Fano” is the 4-graph whose edges are the complements of the seven lines of a Fano plane.

The SHA-256 digest of every file of the data package [30] is listed in its file SHA256SUMS (release v1.0.0), whose own SHA-256 digest is a3a80fb7d2976e6656efb8a27f49d5ac b8d9742dbbc4482581ba0ef637ae503d.

The certificate of Theorem 6 (Section 3.8) has 1 row of W in the block (1, 4), 19 in (3, 5), 4 and 7 in the two types of (4, 5), none in (2, 4) and (3, 4), and 272, 193, 160, 44, 106 in the five types of (5, 6) (with 1, ..., 5 edges), plus the terms $B^\top X B$ of rank 20, 5, 1 on the types with 1, 3, 5 edges; its value is exactly 33/64, and the first, non-sharp certificate has $b = 2267741359833/2^{42} > 0.515624687939$ with $M = 2^{21}$ (class-list scan only; it is not used).

The certificates of Theorem 25 have $Q = A^\top A/M^2$ with $M = 2^{30}$, 2^{19} and 2^{21} for J_4 , K_5^- and K_5^{4-} , and 43, 45 and 10 Gram blocks, one per pair (s, m) and type (the pairs with $s < m$, $m \geq 3$ and $2m - s \leq 7$ for the 3-graphs, those of Section 3.2 for the 4-graphs). The other seven upper bounds of Table 5 come from certificates with $M = 2^{19}$, 2^{19} , 2^{18} , 2^{19} , 2^{21} , 2^{20} , 2^{20} and 45, 46, 40, 48, 12, 12, 12 Gram blocks, in the order of the table, and give

$$\begin{aligned}
\sigma(K_5^-) &\leq \frac{66572174747207}{173173081374720}, & \sigma(K_5^{3-}) &\leq \frac{781068653059}{1924145348608}, & \sigma(C_5) &\leq \frac{2167282889}{8589934592}, \\
\sigma(K_6^{(3)}) &\leq \frac{85699693910797}{115448720916480}, & \sigma(K_6^{4-}) &\leq \frac{565272027609637}{923589767331840}, \\
\sigma(K_6^{(4)}) &\leq \frac{408882571413}{549755813888}, & \sigma(K_7^{(4)}) &\leq \frac{67510962453727}{76965813944320}.
\end{aligned}$$

The exact bounds of Theorem 8 (Table 2) are

$$\begin{aligned}
t_3(5, 4) &\geq \frac{3298534811415}{4398046511104}, & t_3(6, 4) &\geq \frac{11166973764957}{38482906972160}, & t_4(6, 4) &\geq \frac{391547931101}{1099511627776}, \\
t_5(6, 4) &\geq \frac{64873027783677}{153931627888640}, & t_6(6, 4) &\geq \frac{612132591153137}{1231453023109120}, & t_9(6, 4) &\geq \frac{217134104143267}{307863255777280}, \\
t_{11}(6, 4) &\geq \frac{24267758577011}{28862180229120}, & t_2(7, 4) &\geq \frac{111867473425}{1099511627776}, & t_3(7, 4) &\geq \frac{4043518541347}{30786325577728}, \\
&& t_4(7, 4) &\geq \frac{11199502214035}{69269232549888}.
\end{aligned}$$

The exact bounds of Theorem 5 are

$$\begin{aligned}
t(6, 5) &\geq \frac{5410603883828909241569858123831800227}{23261489926236027775816623554906030080} > 0.232599197256, \\
t_3(6, 5) &\geq \frac{28392597905796184739036021237749789573}{4652297985247205551633247109812060160} > 0.610291902965, \\
t(7, 5) &\geq \frac{1006048490611906127582298374928111023}{13956893955741616665489974132943618048} > 0.072082548868,
\end{aligned}$$

where $t_3(6, 5)$ is the least density of 5-graphs in which every six vertices span at least three edges, so that $\pi(H_4^5) = 1 - t_3(6, 5)$.

$T_\lambda(n, p, r), T(n, p, r)$	least number of edges of an r -graph on n vertices in which every p -set spans at least λ edges; $T = T_1$ (Definition 1)
$t_\lambda(p, r), t(p, r)$	$\lim_n T_\lambda(n, p, r) / \binom{n}{r}$; $t = t_1$
$\pi(F)$	Turán density; $\pi(K_p^{(r)}) = 1 - t(p, r)$, $\pi(H_k^r) = 1 - t_{r+2-k}(r+1, r)$, $\pi(K_6^{4-}) = 1 - t_2(6, 4)$
$d(H), c_Q(H), b$	edge density, sum-of-squares term and value of a certificate (Definition 10)
$\text{co}_2(G), \sigma(F)$	codegree-squared sum $\sum_T \deg_G(T)^2$ and codegree-squared density [3]
C_0	$21894/\mu_0 < 7.81 \cdot 10^8$, the constant of Theorem 20
L_v ; locally Giraud	link $\{R : R \cup \{v\} \in E(H)\}$; every 7-vertex induced subgraph is a Giraud system (Section 4)
$\mu_0, q(H), \xi(H)$	margin of the certificate of Theorem 6 outside Giraud’s classes, fraction of non-Giraud 7-sets, and $\kappa(H) - 33/64 + 68/(n-7)$ (Proposition 15)
$\gamma(G), \kappa(H)$	probability that $T \cup \{x\}, T \cup \{y\}$ are both edges; $\kappa = 2d - \gamma$ (Lemma 14)
$L(n, k, r, p), \ell(k, r, p)$	lottery number (tickets of size k , match r , draws of size p) and $\lim_n L(n, k, r, p) / \binom{n}{r}$; the lotto tables write $L(n, k, p, r)$
$\binom{k}{r} \ell(k, r, p)$	normalised lottery constant; the shadow bound says it is at least $t(p, r)$
$\delta(k, t, p)$	Montecalvo’s constant, $\binom{p}{t}$ times the normalised lottery constant
$s(k, r)$	limit density of covering $(k-r)$ -insertion codes; $s(k, r) = t(k, r)$ [36]
$\tau_k(\Gamma)$	fractional cover number of the edges of Γ by k -sets (Section 8.3)

B Notation

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